10.5.5 Magneto Hydro Dynamic (MHD) Generation

In the conventional steam power plants, the heat released by combustion of fuel is transformed into the internal energy of steam. The temperature and pressure of steam increase in the process. The steam turbine, then, converts steam energy into mechanical energy, which drives a generator. This way, the mechanical energy is converted into electric energy. The repeated conversion of various forms of energy involves losses and, hence, the overall efficiency of thermal power plant decreases. The typical range of efficiency of thermal plants is 37 to 40%. The direct conversion of heat to electricity would enable the industry to use the fuel resources more efficiently. MHD generation is one form of energy technology, wherein direct conversion of heat into electric energy has been devised. The technological development in the field of plasma physics and metallurgy etc. and other branches of science and technology has made it possible for this kind of direct transformation of energy.

An ionized gas is used as conducting medium in the MHD generator. The gas can be made electrically conducting when it is maintained at least at a temperature of 2000°C. This fact does not allow MHD generation from being used in the entire temperature range from 3000 K to 300 K. It is, therefore, thought beneficial that MHD generators be used in conjunction with steam operated thermal plants utilising the heat of the gas leaving the MHD ducts. The combined operation of MHD generators alongwith the conventional thermal plant, will raise the overall efficiency to nearly 60%, thereby lot of saving in the fuel cost will result.

Principle of Operation of MHD Generator

The basic principle of operation is based on Faraday's law of electro magnetic induction, which states an e.m.f. is induced in a conductor moving in magnetic field. The conductor may be a solid, liquid or a gaseous one. The study of the dynamics of an electrically conducting fluid interacting with a magnetic field, is called magneto hydro dynamics.

In this method (Fig. 10.12) gases at about 2500°C are passed through the MHD duct across which a strong magnetic field has been applied. Since the gases are hot, and partly ionized they form an electrically conducting conductor moving in the magnetic field. An e.m.f. (direct-current) is thus induced, which can be collected at suitable electrodes.

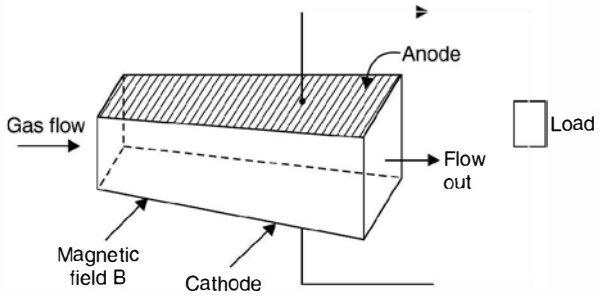


Fig. 10.12. Basic Principle of MHD Generator.

10.5.6 Solar Energy

Sun is the primary source of energy. The earth receives 1.6×10^{18} units of energy from the Sun annually, which is 20,000 times the requirement of mankind on the earth. Some of the solar energy causes evaporation of water, leading to rains and creation of rivers etc. Some of it is

utilized in photosynthesis which is essential for sustenance of life on earth. Man has tried, from time immemorial, to harness this infinite source of energy, but has been able to tap only a negligibly small fraction of this energy, till today.

Three broad categories of possible large scale applications of solar power are :

(*i*) The heating and cooling of residential and commercial buildings;

(*ii*) The chemical and biological conversion of organic material to liquid, solid and gaseous fuels; and

(*iii*) Conversion of solar energy to electricity.

The use of solar energy for generation of electricity is costly as compared to conventional methods. However, due to scarcity of fuel, solar energy will certainly find a place in planning the national energy resources.

10.6 CONVENTIONAL SOURCES

10.6.1 Hydro Station

The water wheel, as developed in the early part of 19th century, played an important role in converting water power into mechanical power. With the invention of steam engines, the use of water wheel began to decrease and larger steam engines were developed. Steam engines possessed the advantage of mobility, allowing power to be produced, where it was required and also that of flexibility in its application.

It was only later with the discovery of conversion of mechanical energy into electric energy, and transmission of electric energy being the most efficient method of transporting energy from one place to another, that water wheel was revived. The modern water turbine, is being built in single unit of more than 200 MW. Also, the concept of multipurpose project, in which the production of power is included as one of several uses (flood control, navigation, irrigation, water for domestic and industrial purpose, etc.), has led to the development of sites which otherwise could not be harnessed economically for power alone. The capital investment per KW is much higher in case of hydro power as compared to thermal power. This is because in order to store water at sufficient head, it is essential to construct a dam which is a costly affair. However, the running cost of hydro electric energy is much less as no fuel is used.

Water power differs fundamentally from thermal power in that it represents an inexhaustible source of energy which is continually replenished by the direct agency of the Sun; whereas thermal power represents chemical energy which has been created and stored within the earth's crust during past geological ages. The use of chemical energy is thus equivalent to the consumption of capital as the replacement is not so easy. Another important difference between the two is that whereas water power can be developed only where it is present in nature, thermal power (liquid or solid fuel) can be transported for use from one place to another.

Classification of Hydro Plants

The hydro-power plants can be classified in terms of location and topographical features, the presence or absence of storage, the range of the operating head etc.

Classification based on Plant Capacity

(*i*) Micro hydel plant	:	less than 5 MW
(*ii*) Medium capacity plant	:	5 to 100 MW
(*iii*) High capacity plant	:	101 MW to 1000 MW.
(*iv*) Super capacity plant	:	above 1000 MW

10.6.2 Steam Power Plant

With the invention of steam engine for obtaining mechanical energy, the so-called non-conventional methods i.e. wind, tidal, geothermal etc. were abandoned as the cost involved was high and also there was no flexibility for transportation of this form of energy. The development of steam turbines and then electric generator completely replaced the non-conventional methods. Fossil fuels became the main source of energy for quite sometime. The size of the thermal plants grew from a few KW to more than 1000 MW as of today. The concept of generating electrical energy using fossil fuel has changed completely, the concept of locating the power plants near the load centres to locating near the fuel pithead. Super-thermal power plants (plants with capacity 1000 MW or more) have come into existance. It has been found more economical in general to generate electrical energy near the pithead rather than near the load centres, even though the energy has to be transported over the transmission lines, which involves a large percentage of total capital cost and transmission line losses. On the other hand by installing a plant near pithead saves the cost of transporting the coal etc. A 400 MW capacity plant requires about 5000 to 6000 tons of coal every day.

10.6.3 Nuclear Power Plants

Nuclear power industry has made relatively faster growth as compared with other forms of power industries. The first power reactor was commissioned sometime in 1944 and by 1972 there were more than 100 nuclear power plants in the world, with the total capacity exceeding 30 GW. The rapid growth can be associated with the following characteristics of the materials used for nuclear power generation :

(*i*) Energy is released without using oxygen for combustion (fission).

(*ii*) Breeding of nuclear fuel is possible so that we produce almost same amount of nuclear fuel as is spent, without reduction of power output.

(*iii*) The weight of fuel required for generating a particular amount of energy is much less than what is required in conventional methods of generation.

Nuclear Fuel

Uranium is the fuel used in nuclear power plants. It is non-renewable and has two isotopes, U'-235 about 0.7% and U-238, 99.3%. Materials fissionable by thermal or low speed neutrons are U-233, U-235 and Pu-239 (plutonium).

As natural Uranium is available in abundance in Britain, most of the reactors are using natural Uranium, whereas United States of America are using enriched Uranium for most of their reactors. In India, Rajasthan Atomic Power Plant uses natural Uranium, whereas Tarapur Power Plant uses enriched Uranium. If natural Uranium is used, the size of the plant is larger than when enriched uranium is used. It has been estimated that about 1.2 kg of Uranium produces 1 MW of power for one year.

10.6.4 The Gas Turbine Plant

The first gas turbine was used in 1939 for large central station service as a prime mover. Since then several stations have been in operation. These plants require less space for the same capacity as compared to a steam plant, it provides more flexibility in design, and involves low initial cost.

Air is used as the working fluid. It is compressed by the compressor and fed to the combustion chamber where it is heated by burning fuel in the chamber or by air heaters. The hot air at high pressure, moves from the combustion chamber to the gas turbine where it expands and delivers mechanical energy. The gas turbine drives the alternator which converts mechanical energy into electrical energy. The compressor, the gas turbine and the alternator are mounted on the same shaft so that a part of turbine power is supplied to the compressor besides the alternator. Therefore, the efficiency of the gas plant is relatively low (about 20%).

Comparison of Salient Features of Various Types of Power Plants

S.No.	*Particulars*	*Hydro plant*	*Steam plant*	*Nuclear plant*	*Gas plant*
1.	Site	Large quantities of water at sufficient head. Feasibility of constructing economical dam.	Normally located near the load centre. Superthermal Power Plants at the coal pit head.	Normally located near the load centre, away from the thickly populated areas to avoid radioactive pollution.	Can be installed anywhere.
2.	Initial cost	Highest because of civil works	Low	In between Hydro and Steam	Lowest
3.	Running cost	Lowest	In between hydro and nuclear	Highest	Lower than nuclear
4.	Maintenance Cost	Lowest	In between hydro and nuclear	Highest	Lower than steam and nuclear
5.	Transmission and Distribution cost	Highest	Low	Lower	Lowest
6.	Simplicity and cleanliness	Simple and clean	Causes air pollution	Equipment handling complex, leakage may lead to disastrous air pollution.	Simpler and cleaner than steam.
7.	Overall efficiency	Most efficient, approx, 85%	Least efficient. Overall approx. 25%	Quite efficient approx. 30%	More than diesel, approx. 40%
8.	Starting	Instant starting	Max. time (about 1/2 hr.)	Time less than for steam plant.	About 2 min.
9.	Space requirements.	Maximum	Less than that for Hydro	Least	Less than steam
10.	Field of application	Peak or base load	Base load	base load	Peak load or as standby unit.
11.	Reliability	Most reliable	Less reliable	Reliable	Reliable

PROBLEMS

10.1. Draw single line diagram of a power system. Label all the major components of the system.

10.2. Discuss the advantages of ring main distribution system over the radial system.

10.3. Compare a 3-phase 4-wire system with the single phase system.

10.4. Describe briefly various supply systems in power system.

10.5. What are the causes of low p.f. in power system and discuss various methods of improving the power factor.

10.6. Compare the performance of static capacitor with a synchronous condenser as a p.f. improvement device.

10.7. A synchronous generator supplying 2000 kW operates in parallel with an induction generator supplying 1000 kW. The load has a lagging p.f. of 0.8. The induction generator has a p.f. of 0.95. Determine the p.f. at which the synchronous machine is working. [**Ans.** 0.612]

10.8. A synchronous motor overexcited to take 600 kVA gives 360 hp at an efficiency of 0.9. It runs in parallel with a load of 1000 kVA at p.f. 0.8 lag. Determine the resultant p.f. [**Ans.** 0.99 lag]

10.9. Briefly discuss various non-conventional methods of generating electrical energy.

10.10. "The non-conventional methods were known to man since time immemorial, still they were not developed". Comment and give reasons.

10.11. Explain why it is necessary to develop non-conventional methods of generating electrical energy.

10.12. Discuss the advantages of tidal power and name a few projects, harnessing tidal power.

CHAPTER

11 Domestic Wiring

11.1 INTRODUCTION

The practice of domestic wiring has been standardised. Any deviation for these practices may lead to untoward incident like short circuits or electric shock to personnel *e.g.* it is mandatory to place a fuse wire in the phase wire (live wire). An attempt to put it in the natural may create problem after the fuse blows off due to short circuit. Proper material should be used for laying the domestic wiring depending upon the environment, location, safety to personnel, to equipments etc.

11.2 TYPES OF DOMESTIC WIRING

Following are the different types of domestic wiring :

1. Cleat wiring
2. Wooden casing and capping wiring
3. TRS or Batton wiring
4. Surface conduit wiring
5. Concealed conduit wiring

The above listing is in the direction of increasing cost ; temporary/permanent wiring and better appearance. We discuss them in brief as follows :

11.2.1 Cleat Wiring

In this system of wiring the cables are gripped and supported between cleat made of porcelain. These cleats are made in two halves; the lower part is grooved to accommodate, the cable and the upper part is put over it. After the cable is put-between the cleats, these are then screwed on wooden plugs also known as gutties. The gutties should first be fixed into the wall at regular intervals of about 50 cm. The screws used are of size 38 mm length. For 250 volts supply, the cables will be placed 2.5 cm apart centre to centre for single core cables. Only one cable is passed through each groove of the cleat. While screwing too much pressure should not be applied so that its insulation remains intact. The cables normally used in cleat wiring are VIR (Vulcanised India rubber) cable, PVC cables etc. In this system of wiring sharp bends and undue twisting should be avoided. Also, whenever cable has to pass through walls, floors or any other concealed location, the conduit pipe of suitable length should be used for that portion of wiring. Cleats for two wires are shown in Fig. 11.1.

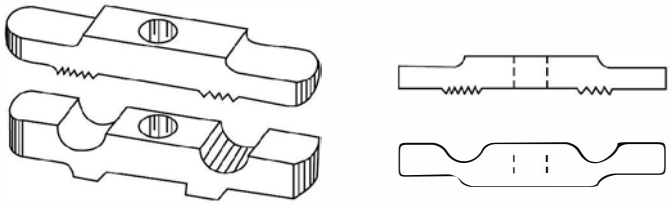


Fig. 11.1 Cleats for two wires.

The advantages and disadvantages of cleat wiring are:

Advantages:

1. The installation and dismentling is simple and fast and hence is very useful for temporary wiring.
2. The cable can be inspected easily and if there is any problem with the wiring the same can be rectified without much problem.

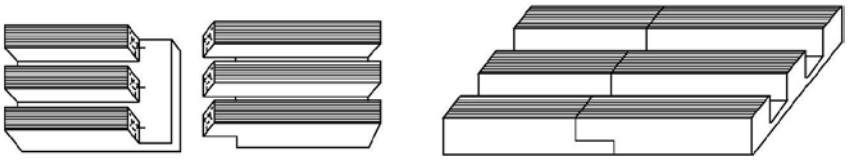
Disadvantages:

1. The appearance of the wiring is not good.
2. It cannot be used in damp places.
3. As the wiring is exposed to atmosphere, its life span is short.

11.2.2 Wooden/PVC Casing and Capping Wiring

This method of wiring is one of the very old method and is in use even as of now. This system is suitable for the voltages installations using VIR or PVC cables.

The Casing and Capping form two parts of the wooden or PVC, the Casing and Capping have grooves. The cable is placed in the grooves of the Casing and this is then covered by capping as shown in Fig. 11.2. The Capping is fitted to the casing with the help of screws of 12.7 mm length. The Casing and Capping should be made of teak wood.



Two casing shaped for straight joint A finished straight joint

Fig. 11.2 Two casing shaped for straight joint.

The casing should be spaced from the wall by means of small porcelain blocks or disc insulators not less than 6.5 mm thick so that direct contact to the wall is avoided and hence it is protected against dampness of wall.

The following are the advantages and disadvantages of this method of wiring.

Advantages:

1. It is relatively cheaper as compared to conduit but costlier as compared to cleat-wiring.

2. Short-circuit of wire *i.e.* between phase and neutral is avoided as the two are placed in different grooves.
3. Physical inspection of wiring makes it simple to carry out any repair if required (by opening the capping).
4. PVC casing capping gives better look and is economical as compared to wood.

Disadvantages :

1. In case of a short-circuit, there is risk of fire (use of wood or PVC).
2. Normally not recommended for damp places.

11.2.3 Toughened Rubber Sheath (TRS or CTS) or Batton Wiring

In this case the cables are carried on seasoned teak wood perfectly straight and well varnished teak wood batton of thickness not less than 1 cm. The cables are fixed on the batton by means of tinned brass or aluminium link clips already fixed on the batton with small nails before laying cables. The battons are fixed to the walls by means of gutties with countersink headed wooden screws. The screws are to be fixed on the batton at an interval of about 75 cms. The minimum width of batton is 13 mm for two wires. It is found suitable where acids and alkalies are stored. The various sizes of batton available are 13, 19, 25, 31, 38, 44 and 50 mm width. Depending upon the no. of cables to be run, a suitable size of batton should be used.

The main advantages and disadvantages of this wiring are :

1. Easy to instal and economical as compared to casing cappings or conduit wiring.
2. Easy to detect fault if any and has good appearance.
3. Relatively good life span.

Disadvantages :

1. Not suitable if exposed to sun and rain or places where dampness exists.
2. Risk of fire high.

CTS stands for Cab Type sheathed.

11.2.4 Conduit Wiring

There are three types of conduit wiring :

(*a*) Concealed conduit wiring

(*b*) Surface conduit wiring

(*c*) Flexible conduit wiring.

Concealed conduit wiring. Here the conduits are embedded along the wall in plaster at the time of construction. The VIR or PVC cables are drawn into the conduits by means of GI wire of 18 SWG. The conduit should be electrically and mechanically continuous and connected to earth at some suitable places using earth wire. This method of conduit wiring is preferred for domestic over other methods as it maintains the beauty of the house and no projected pipes are visible.

Surface conduit wiring. As the name suggests the conduit is spaced from the wall by means of small wooden spacers below the conduits along its length at regular intervals. This system is commonly used in industrial wiring but is not recommended for domestic wiring as it spoils the beauty of the house. The usual size of the conduit is 25 mm diameter for domestic wiring.

Flexible conduit wiring. This method of wiring is used to provide mechanical protection to cables between rigid conduits (surface conduit) and electric machine or any other electrical device. However, this is not a general method of wiring. It is used for connecting rigid conduit with machine terminal box in case of motor wiring and energy meter with main switch in case of industrial and domestic wiring.

The following are the advantages and disadvantages of conduit wiring :

Advantages :

1. It provides protection against mechanical injury to insulated cables and against short-circuits.
2. Since conduits can be made water tight, the conduit wiring provides protection against moisture.
3. Gives better appearance.
4. Relatively longer life.

Disadvantages :

1. It is a relatively costlier system of wiring.
2. Highly skilled workmanship is required and it is more time consuming.

The method of wiring is normally recommended where cost is not the consideration *e.g.* in big hotels, workshops and factories and places of storing inflammable materials.

11.3 SPECIFICATIONS OF WIRES

Usually cables or wires have a no. of strands twisted together. Standard stranding consists of 6 wires around 1, then 12 around 6, followed by 18, 24 and so on. A stranded conductor is expressed as 7/0.1 where the first number stands for the no. of strands used and the second number corresponds to the diameter of each strand in mm. Some times the second number given corresponds to the gauge of the strand used *e.g.*, 3/20 which means a cable with three strands each of 20SWG.

All electric cables consist of three essential points :

(*a*) the conductor for transmitting electrical power,

(*b*) the insulation, an electrical insulating medium, needed to insulate the conductor from direct contact with earth or other objects, and

(*c*) external protection against mechanical damage, chemical or electro-chemical attack, fire or any other dangerous effects external to the cable.

Copper conductor has extensively been used for cables but of late aluminium is being used to a considerable extent. To obtain flexibility a number of wires are made up into a strand which makes it easier to handle, less liable to kink and break and to a large extent eliminates risk to the conductor breaking through the dielectric. The wires in a stranded conductor are twisted together to form lays. The successive layers usually are stranded in opposite direction *i.e.*, if the wires of one layer have a right-handed lay, the next layer has a left-handed lay.

In regard to specification of cables, it is important to mention all possible details regarding the particular type of cable in terms of size (in mm or mm^2) types of conductor whether copper or aluminium, type of insulation PVC or VIR and voltage grading.

11.3.1 Size of Conductor

Following points must be kept in mind while determining the size of the conductor for internal wiring for a given circuit.

(*i*) Minimum size of the conductor for mechanical reason

(*ii*) Current carrying capacity

(*iii*) Voltage drop

Minimum size of Conductor

(*i*) **Wire.** The minimum size of copper conductor in use is 1/1.2 mm based on permissible wattage of 800 watts in the subcircuit. The aluminium conductor of size 1/1.4 mm is used for a subcircuit in domestic wiring and minimum size of conductor for power wiring is 4 mm^2 and 1/2.24 mm.

(*ii*) **Underground cables.** The area of cross section of conductor for two core cables should not be less than 6 mm^2 and for three and four core cables it should not be less than 25 mm^2. The minimum area of cross section of a three and half core cable should be 50 sq. cm.

Current-carrying capacity. When current is passed through a wire a certain amount of power is wasted in the wire which appears in the form of heat which results in increase in temperature of the conductor which may again decrease the life of insulation and, therefore, the rating should be such that when that current is allowed to flow continuously it does not affect the insulation of the cable. However, it should be able to carry a relatively larger current. (Varying between 1.5 to 2 times the continuous rated current.) for a short duration without affecting the insulation.

Voltage drop. Before deciding a proper size of cable to be used in a circuit, due consideration must be given to voltage drop. This voltage drop in the cable from distribution board to the point of device (lamp, fan, motor etc.) is due to the flow of current through the cable. The permissible voltage drop from supply terminal to any point on the wiring system should not exceed 2% + 1 volt for light loads and 7.5% for declared supply voltage of power loads.

We take a few examples to illustrate the selection of specification for wires:

Example 11.1. *Determine the size of the conductor for power and lighting circuit from meter to main distribution board having three light/fan circuits of 800 watt each and two 15A circuits (power circuits) of 1500 watt each. Assume negligible voltage drop.*

Solution. Load wattage in 5 A circuit $= 3 \times 800 = 2400$ watts.

Load wattage in 15 A circuit $= 2 \times 1500 = 3000$ watts.

Total load $= 5400$ watts.

Assuming an average p.f. of 0.8

Current in amperes at 230 volt supply.

$$\frac{5400}{230 \times 0.8} = 29.3 \text{ A} \simeq 30 \text{ A}$$

Hence current under short circuit $30 \times 1.5 = 45$ A

From the table 11.1, for aluminium wire the size of the conductor comes out to be 25 mm^2. In fact for 43 A it is 16 mm^2 but we should always go for one higher size of the conductor and hence we select conductor of size 25 mm^2 or 7/2.24 mm.

Table 11.1. Current Rating of Aluminium conductor Single core cables for VIR, PVC or Polythene insulated including tough rubber, PVC

Size of conductor		*2 cables d.c. or single phase a.c.*		*3 or 4 cables or balanced three phase*		*Four cables d.c. or single phase a.c.*	
Nomial area in Sq. mm	*No. and diameter of wire in mm.*	*Current rating in Amperes*	*Approx. length of run for one volt. (M.)*	*Current rating in Amps.*	*Approx. run for one volt drop in (Mt.)*	*Current rating (amps.)*	*Approx. run for one volt drop in Mts.*
1.5	1/1.40	10	2.3	9	2.9	9	2.5
2.5	1/1.80	15	2.5	12	3.6	11	3.4
4.0	1/2.24	20	2.9	17	3.9	15	4.4
6.0	1/2.80	27	3.4	24	4.3	21	4.3
10.0	1/3.55	34	4.3	31	5.4	27	5.4
16.0	7/1.70	43	5.4	38	7.0	35	6.8
25.0	7/2.24	59	6.8	54	8.5	48	8.5
35.0	7/2.50	69	7.2	62	9.3	55	9.0
50.0	7/3.50 19/1.80	91	7.9	82	10.1	69	10.0
70.0	19/2.24	134	8.0	131	9.5	—	—
95.5	19/2.50	153	8.8	152	10.1	—	—

Example 11.2. *A single phase supply is to be brought to a residence from the mains which is at a distance of 25 m. The supply is at 230 V and the current to be carried is 5 A, determine the size of the conductor.*

Solution. Permissible voltage drop = 1 + 2% of 230 V

= 1 + 4.6 = 5.6 volts

Referring to the table for selection of wire of aluminium conductor, the min. size of wire is of 1.5 mm^2 section or 10 A and produces a voltage drop of 1V for using 2.3 metre length. Hence voltage drop at 10A is $\frac{25}{2.3}$ volts

Hence voltage drop at 5 A $= \frac{25}{2.3} \times \frac{5}{10}$

= 5.4 volt

As the permissible drop is 5.6 volts and the conductor with 1.5 mm^2 section gives voltage drop of 5.4 volts hence the suitable size is 1/1.40 mm. **Ans.**

Example 11.3. *Determine the size of the conductor to be used for wiring a 10 kW 400 V, 3-phase 50 Hz induction motor. The voltage drop is negligible between the main switch and the motor terminals.*

Solution. Assuming efficiency of 85% and a p.f. of 0.8 the input current to the motor is

$$\frac{10 \times 1000}{\sqrt{3} \times 400 \times 0.8 \times 0.85} = 21.2 \text{ A}$$

At the time of starting the induction motor may take 2 times the rated current and hence starting current = 21.2 x 2.0 = 42.4 A.

From the table corresponding to 42.4 A *i.e.* 43 A the wire used is 7/1.7 mm or 16 mm^2.

11.4 DISTRIBUTION BOARD

Fig. 11.3 shows a distribution board located in covered Verandah in a house or in the room where there is no Verandah. Care should be taken to see that privacy should not be disturbed while selecting the location for meter board as the meter reader shall be visiting the house regularly.

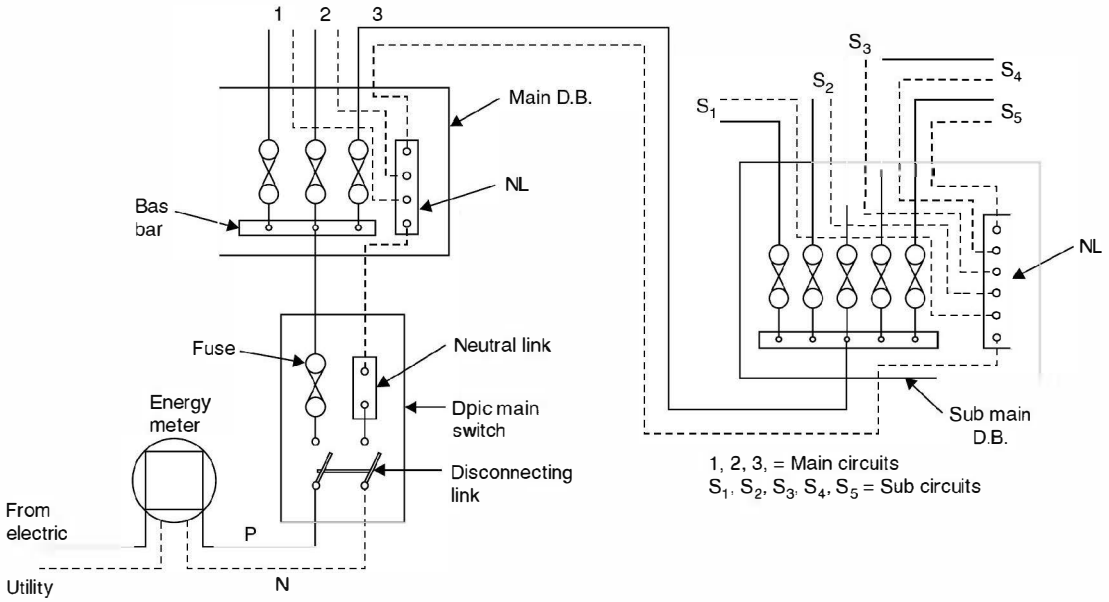


Fig. 11.3 Distribution Board.

The cable from the electric utility is connected to the energy meter first before taking it to any other connection. From the load terminals of energy meter it is taken to distribution through a double pole iron clad switch (DPIC). The phase wire is connected to a metal bar or bus bar of the main distribution board through a fuse, and neutral link of the DPIC switch is connected to neutral link of the main distribution board. From here the neutral wire is distributed one each with various phase wires to form a complete sub circuit. It is important to note that neutral wire is never looped from one circuit to the other.

11.5 TYPES OF CABLES

A brief description of the various cables used in domestic wiring is given here.

The main requirements of the insulting materials used for cable are :

1. High insulation resistance.
2. High dielectric strength.
3. Good mechanical properties *i.e.* tenacity and elasticity.
4. It should not be affected by chemicals around it.
5. It should be non-hygroscopic because the dielectric strength of any material goes very much down with moisture connect.

Vulcanized Rubber

Rubber in its natural form is highly insulating but it absorbs moisture readily and gets oxidized into a resinous material; thereby it loses insulating properties. When it is mixed with sulphur alongwith other carefully chosen ingredients and is subjected to a particular temperature it changes into vulcanized rubber which does not absorb moisture and has better insulating properties than even the pure rubber. It is elastic and resilient.

The electrical properties expected of rubber insulation are high break-down strength and high insulation resistance. In fact the insulation strength of the vulcanized rubber is so good that for lower voltages the radial thickness is limited due to mechanical consideration.

The physical properties expected of rubber insulation are that the cable should withstand normal hazards of installation and it should give trouble-free service.

Vulcanized rubber insulated cables are used for wiring of houses, building and factories for low power work.

PVC (Polyvinyl chloride) cables. These are thermoplastic insulating materials and not used for high temperature as it gets softened and flows down to heat. These are therefore, not used for heating appliances. PVC is harder than rubber, hence a thin layer of PVC insulation is good enough. In fact, its thickness is decided by mechanical reasons rather than electrical. The PVC insulated cables are lesser in diameter as compared to TRS and more number of wires can be placed in conduit as compared to TRS wires. These are used upto 1.1 kV voltages especially in concealed wiring system.

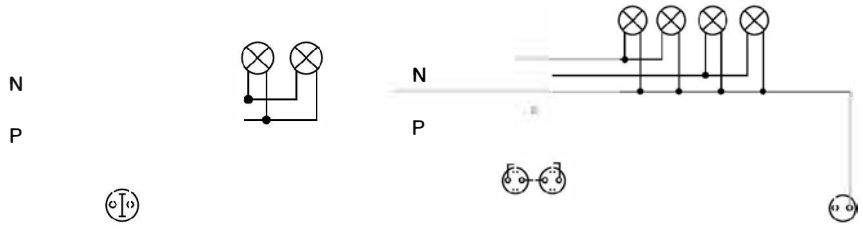
Tough Rubber Sheathed (TRS) or Cab Type Sheathed (CTS) Cables. This cable is used where humidity is high and it does not deteriorate even during long duration of exposure to moisture. As the name suggests these cables are provided with outer protective covering of tough rubber over normal insulation. These are available as single, twin and three cores with an earth continuity conductor. These are used on 250/500 volt circuits.

11.6 LIGHTING CONTROL CIRCUITS

Various lighting control circuits have been developed for maximum convenience and economical use of electric lighting.

Fig. 11.4 (*a*) shows a circuit diagram for switching two incandescent lamps with a single pole lighting switch. The no. of lamps in the circuit may be one, two or more. Fig. 11.4 (*b*) shows

a circuit diagram controlling several lamps with two lighting switches, the circuit having one plug socket at its end. Under all circumstances the plug socket must be independent of the operation of the general lighting units and it must be connected to the nearest point in the common supply circuit.



(*a*) Circuit for lamp control with a single-pole lighting switch.

(*b*) Circuit controlling several lamps with two or more lighting switches.

Fig. 11.4

Fig. 11.5 (*a*) shows circuit for control of three lamps in one room by a chandelier multi way switch. As can be seen that the first turn of the switch lights one lamp, the second turn lights two lamps, but the first lamp goes out. The third turn will light up all the three lamps, and when the fourth turn is made, all the lamps go out. The circuit diagram also shows that a SOCKET is connected in the circuit near the multi-way switch.

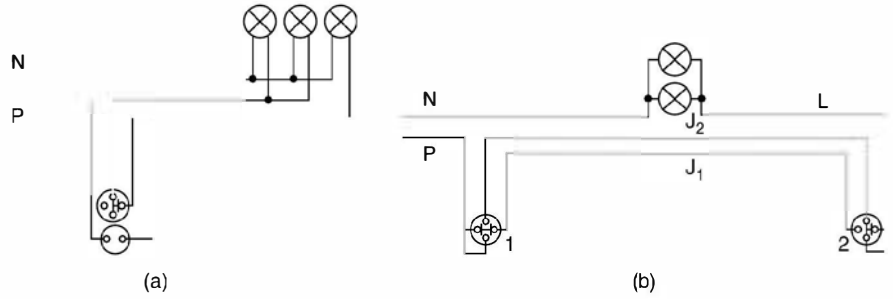
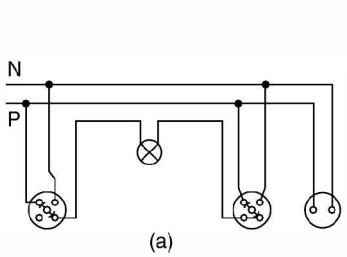


Fig. 11.5 Circuit for controlling a group of lamps from two different places.

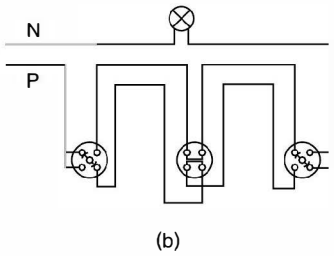
Fig. 11.5 (*b*) shows a circuit by which independent control of lamps from two different places (Stair Case Wiring) is carried out with the help of corridor switches. It can be seen that corridor switch 1 transfers the supply from jumper J_1 to jumper J_2 and that corridor switch 2 performs the same operation with lamp conductor L. This makes the operation of the switches independent of each other, as each of them, in breaking the circuit, simultaneously prepares it for closing by the other switch.

Fig. 11.6 (*a*) shows the same circuit with a plug socket provided at the end of the circuit whereas Fig. 11.6 (*b*) shows a circuit used to control lamps from several places.



(a)

(*a*) Circuit for controlling a group of lamps from two different places and also having a plug socket installed at its end.



(b)

(*b*) Circuit for controlling a group of lamps from several places.

Fig. 11.6

The above circuits take in all main ways of controlling lighting installations. It is only possible to change the number of lamps and switches, change their positions relative to one another, include one or another number of plug sockets and arrange them at different places.

11.7 EARTHING SYSTEM

By earthing of electrical installations is meant the direct connection of all the metal non current carrying parts of electrical equipments *e.g.,* metallic framework, body of electric machines, main switch, metallic covering of cables and conduits, distribution board, earth terminal of sockets etc. to earth plate. The earth plate is buried in the ground which has good electrical connection to the surrounding earth. As far as domestic wiring is concerned the electric utility provides earthing to its energy meter. But beyond the meter, earthing is the responsibility of the owner of the house. The owner should make arrangements of his own earthing system with an independent electrode. All appliances which are given connection through the socket outlet *e.g.,* refrigerator, washing machine, drier machine, table fan etc. should be earthed through earth terminal of the socket outlet.

A safe grounding system should provide the following :

(*i*) A means to carry and dissipate electric currents into ground under normal and abnormal operating conditions so that not only the equipment should remain safe, the continuity of supply should also be maintained.

(*ii*) The personnel working on or around the equipment should not experience any electric shock.

We will discuss here in detail the second aspect of grounding *i.e.,* personnel safety. For this it is important to understand the electrical characteristics of the most important part of the circuit *i.e.,* the human body.

It is to be noted that human body experiences shock due to flow of current through the body and not due to the voltage level. If the human body is properly insulated he or she can even hold bare high voltage wire. It is a usual sight to see birds sitting on the wire of h.t. lines.

However, if the birds span the two HT wire simultaneously, it will turn into a dead short circuit. In general, shock currents are classified based on severity of the shock they cause *e.g.,* currents which produce direct physiological harm are termed primary shock currents, whereas,

those that do not cause direct physiological but may cause involuntary muscular reactions are called secondary shock currents. It is to be noted that the threshold value of current which gives a tingling sensation on the hand or finger when touching an electrically live device differs from person to person. For a normal healthy person a current of 1 mA is the threshold value to produce tingling sensation. Currents of about 10 to 30 mA can cause lack of muscular control.

In most of the cases a current of 100 mA will cause ventricular fibrillation. Currents of higher magnitude may even completely stop the functioning of heart or may cause severe electrical burns. Ventricular fibrillation is that condition when the heart beats in an abnormal and ineffective manner with fatal consequences. Therefore, the threshold value of current is the main criterion for proper grounding of devices. Under threshold condition, there is rapid uncoordinated contractions of the ventricles of the heart thereby the synchronisation between the heart beat and pulse beat is lost. When individual muscle fibres contract independently, they are said to be fibrillating. Fibrillation can occur in the atrial or ventricular muscles. During ventricular fibrillation the *ECG* (Electrocardiogram) is totally disorganised.

A human heart can be considered as a muscle operating rhythmically due to a nerve pulse that provides the heart beat. Therefore, when an external signal due to electric current is sent into the heart which will have different frequency, different from that of the normal heart, it disturbs the rhythmic flow of operation of heart. This condition of operation is known as ventricular fibrillation or arrhythmic operation of heart. Once this arrhythmic condition is set up, it is difficult to stop. It usually requires injection of another electric current to stop the fibrillation and reestablish the normal rhythm. Table 11.2 gives typical effects of Electric shocks current on humans.

Table 11.2. Typical effects of Electric shock currents on humans

50 Hz current	*Effect*
0–1 mA	No sensation
0–3 mA	Mild sensation
3–5 mA	Pain or annoyance
5–10 mA	Painful shock
10–15 mA	Local muscle contraction
30–50 mA	Breathing difficult, can cause unconsciousness
50–100 mA	Possible ventricular fibrillation of heart
100–200 mA	Ventricular fibrillation of heart certain.

It is to be noted that the effect of electric shock currents not only depends upon the physiological features of a person but the psychologic factors also play an important part. The effects of electric current on vital human organs not only depend upon the magnitude of currents but it depends upon the duration and frequency of the current. Humans are more vulnerable to electric shock current at 50–60 Hz. The human body can withstand slightly higher current at 25 Hz and almost five times larger for d.c. current. Similarly at frequencies of 1 KHz or 10 KHz even larger currents can be tolerated. In case of lightning (where the frequency is very high and duration is in μ secs) the human body can withstand very high currents in terms of several hundreds of amperes.

When the human body becomes a part of the electric circuit the current that passes through it, is limited by the body resistance and the resistance of the earthing system of the device to which the person comes in contact. The human body resistance differs from person to person and also it is different when two feet or between one hand and one feet or between two hands. Generally two types of potentials are important from electric shock point of view. These are

(*i*) Touch potential and

(*ii*) Step potential

When a person touches the electrically live part of a device, the potential between the hand and the foot is known as touch potential. However the potential between the two feet of the person when walking on an electrically conducting and live surface is known as step potential. Usually, the step potentials are smaller in magnitude as compared to the touch potentials.

For grounding a device, a galvanised steel wire is connected to it's body and the other end of the wire is connected to a steel rod which is put into the ground upto a certain depth. The pit contains moisturised mixture of salt and charcoal to provide better conductivity. The steel rod or any other electrode is placed in this mixture. Sometimes more than one rod are used to improve the grounding *i.e.,* reduce the earthing resistance. The earthing resistance also depends upon the resistivity of the soil. In transmission lines, tower footing resistance is reduced by using radial or parallel galvanised steel wires in the ground. These are known as counterpoise wires. The acceptable value of grounding resistance varies upon its application *e.g.,* for domestic appliances, the acceptable value is about 1 ohm and for substation equipment a 5 ohm resistance is acceptable. It is seen that the thickness of the rod does not play a major role in reducing the ground resistance as does the length of the rod. Therefore, it is better to use thin but long rods or many small rods.

Some of the safety measures to avoid electric shock are given below :

1. See that all the metallic parts of various equipments and appliances are properly earthed.
2. Before energising the domestic wiring its insulation must be checked.
3. The phase wire must be connected through the switch.
4. Never try to handle or operate any electrical appliances with wet hands or standing on a wet floor.
5. Make it a habit to switch off the supply and then pull out the plug whenever you have finished with a particular gadget.

11.8 FUSES AND HRC FUSES

A fuse is a small piece of wire connected in between two terminals mounted on insulated base and is connected in series with the circuit. The fuse is perhaps the cheapest and simplest form of protection and is used for protecting low voltage equipments against overloads and/or short circuits. The fuse is expected to carry the normal working current safely without overheating and during overloads or short circuits it gets heated up to melting point rapidly. The materials used normally are tin, lead, silver, zinc, aluminium, copper etc. For small values of currents an alloy of lead and tin in the ratio of 37 per cent and 63 per cent respectively is used. For currents more than 15 amperes this alloy is not used as the diameter of the wire will be large and after fusing, the metal released will be excessive. Silver is found to be quite satisfactory as a fuse material because it is

not subjected to oxidation and its oxide is unstable. The only drawback is that it is a relatively costlier material. Therefore, for low range current circuits either lead-tin alloy or copper is used.

Definition

Fuse. Fuse is a device used in circuit for protecting electrical equipments against overloads and/or short circuits.

Fuse element or fuse wire. It is that part of the fuse device which melts when an excessive current flows in the circuit and thus isolates the faulty device from the supply circuit.

Minimum fusing current. Minimum fusing current is a value corresponding to operation in an arbitrary time obtained under prescribed test conditions. Alternatively, it is a value of current corresponding to a chosen value of time indicated on a time/current curve which is itself obtained from prescribed testing condition.

Fuse rating. It is that value of current which when flows through the element, does not melt it. This value of the fuse is less than the minimum fusing current.

Fusing factor. This is the ratio of minimum fusing current to the current rating of fusing element, *i.e.*

$$\text{Fusing factor} = \frac{\text{Minimum fusing current}}{\text{fuse rating}}$$

The fusing factor is always greater than unity.

Prospective current. It is defined as the rms value of current which would flow in a circuit immediately following the fuse when a short circuit occurs assuming that the fuse has been replaced by a link of negligible resistance.

Melting time or pre-arcing time. The time taken from the instant the current that causes a break in the fuse wire starts flowing, to the instant the arc is initiated.

Arcing time. The time taken from the instant of arc initiation to the instant of arc being extinguished.

Total operating time. It is the sum of the pre-arcing and the arcing time.

The most commonly used fuse in 'house wiring' and small current circuits is the semi-enclosed or rewirable fuse. Whenever the fuse wire blows off due to overload or short-circuit, the fuse carrier can be pulled out, the new wire can be placed and the supply can be restored. This looks simple and is really very simple only if the wire is replaced by the correct size. For a layman this may prove dangerous if he replaces the fuse wire by some copper wire not to the specification and there is a possibility of burning the equipment. Besides, the fuse wire, since is exposed to atmosphere, it is affected by ambient temperature. The time-current characteristics of such fuses get deteriorated with time and hence are not reliable for discrimination purposes. These fuses are, therefore, mainly used for domestic and lighting loads. For all important and costly equipments operating at low voltages (upto 33 kV) another class of fuse is used which is known as cartridge fuse. These are described below. When the HRC (High Rupturing Capacity) cartridge fuse link was first introduced, it was designed to satisfy two important requirements. The first was to cope up with the increasing rupturing capacity on the supply system and the second was non-deterioration to overcome the serious disadvantages suffered by the types of semi-enclosed fuses.

Construction of HRC fuse

The HRC fuse consists of a ceramic body usually of steatite, pure silver element, clean silica quartz, asbestos washers, porcelain plugs, brass endcaps and copper tags (see Fig. 11.7). The brass end-caps and copper tags are electro-tinned. The metal end-caps are screwed to the ceramic body by means of special forged screws to withstand the pressure developed under short

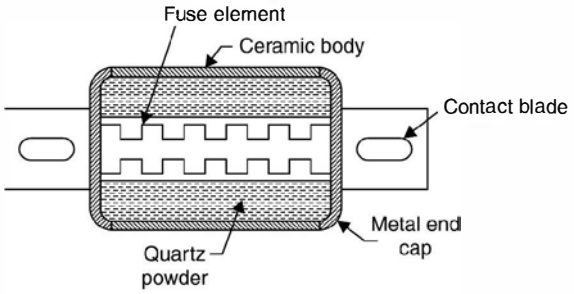


Fig. 11.7 HRC fuse link.

circuit condition. The contacts are welded to the end-caps. The assembly also includes solder of various types, cement and indicator devices. Deterioration of the fuse must involve a change in one or more of these meterials or a change in their structure. Normally the fuse element has two or more sections joined by means of a tin joint. The fuse wire is not in the form of a long cylindrical wire as after it melts, it will form a string of droplets and will result into an arc between the droplets. Afterwards these droplets will also evaporate and a long arc will be struck. The purpose of the tin joint is to limit the temperature of fuse under small overload conditions. The melting point of silver is 960°C while that of pure tin is 230°C. As the circuit is overloaded the melting of tin prevents the silver element from attaining high temperature. The shape of the fuse element depends upon the time-current characteristic required.

Fuse operation

When an HRC fuse operates, the element absorbs energy from the circuit and heats until it melts. The heat produced during operation is given by $\int i^2 R\,dt$ where R is the instantaneous resistance and i the instantaneous current during the operating time. The fuse element melts before the fault current reaches its peak value. As the element melts, it vaporizes and disperses. This action is then followed by a period of arcing during which chemical reaction between the silver vapour and the quartz powder takes place, which further results into building up a high resistance and reduces the current to zero. Thus the arc is quenched. Generally, the filling powder used is quartz-sand as it can absorb heat at a very high rate and it does not evolve appreciable amount of gas.

Cut-off Current

When an HRC fuse interrupts a heavy fault it exhibits an ability to limit the short circuit current. This ability is referred to as a 'cut off' as shown in Fig. 11.8 and has the effect of reducing the magnetic and thermal stresses both in the system and within the fuse itself under fault conditions. Cut-off is in fact one of the main reasons why HRC fuse is so successful as a protective device and it is at times preferred over the circuit breaker of low ratings. Due to this property of

the HRC fuse the operating time is as low as 1/4th of a cycle. The maximum to which the fault current reaches before the fuse melts is called the cut-off current.

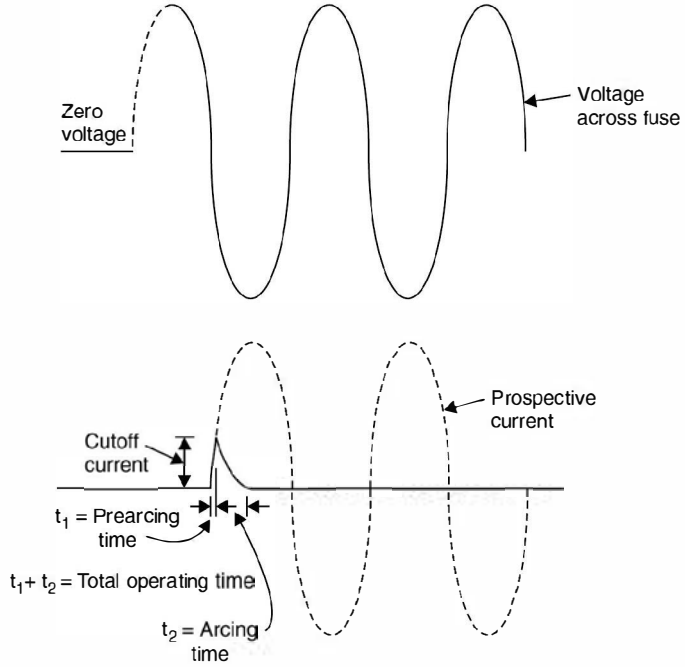


Fig. 11.8 Cut-off characteristic of HRC fuse.

Properties of HRC Fuses

The careful designs of HRC fuses have overcome the disadvantages of the conventional rewirable fuses and their properties are described as follows :

(*i*) *Fast operation.* The HRC fuse interrupts the short circuit current long before its maximum value is attained which is not true in case of CBs. This property of the HRC fuse reduces both thermal and magnetic stresses on the equipment to be protected and the short circuit fault is interrupted well within the first quarter of a cycle.

(*ii*) *Rupturing capacity.* The rupturing capacity of a fuse is expressed in terms of MVA and is equal to the product of service voltage and the rms value of the prospective current it can handle. Since the fuse melts much before the current reaches prospective value due to its cut-off property, it is clear that a fuse is never called upon to carry a current equal to its rupturing capacity. It is to be noted here that the rupturing capacity of a breaker is different from the rupturing capacity of a fuse.

(*iii*) *Non-deterioration.* This means that all the characteristics of the fuse are maintained throughout its life. As the fuses are called upon to function only once in a while, it is most essential that they should preserve their characteristics throughout their useful life. This also implies that they should not operate inadvertently when carrying normal load currents as so often happens with a rewirable fuse which may fail due to oxidation and reduction of cross-sectional area. This property of HRC fuses is very important and is achieved by the hermetic

sealing of the silver element within the fuse body with the help of special cementing and the soldering of the end caps. It has been found that HRC fuses maintain non-deterioration property unimpaired even after approximately 20 years of their manufacture.

(*iv*) *Low-temperature operation.* This is required to eliminate the deterioration of the fuses and to prevent overheating of associated contacts. This is achieved by employing fabricated elements of pure silver which are specially designed to give a low temperature rise when carrying their full-rated current.

(*v*) *Accurate discrimination.* By this characteristic is meant that an HRC fuse on a distribution system will isolate the faulty section from the healthy section whenever a fault takes place. In case of an HRC fuse it is found that the time of operation is inversely proportional to the prospective short circuit current over a much wider range of fault condition and, within practical limits, while the values of prospective short circuit current increase, the time of operation will continue to decrease without reaching a definite minimum. This means that a fuse of low current rating will blow before a fuse of a higher rating, no matter how heavy the fault. It is, therefore, desirable while designing the installation from the view point of discrimination to use fuses of the same design and characteristics throughout, which will ensure that time-current characteristics of each succeeding current size will not cross and the characteristics will be parallel to one another upto the maximum values of fault current.

Arc voltage within safe limits. Whenever an inductive circuit is interrupted, there is likelihood of large voltages induced. The magnitude of such voltages depend upon the magnitude of the short circuit to be interrupted and the circuit constants. A careful design of the HRC fuse controls these over voltages and keeps them within safe limits.

Low cost. It is known that because of the cut-off characteristics of the HRC fuse, for the same rupturing capacity the actual current to be interrupted by an HRC fuse is much less as compared to any other interrupting device and hence it is less expensive as compared to other interrupting devices. It is, therefore, usual to employ a circuit breaker of low rupturing capacity backed up by an HRC fuse where circuit breakers are necessary for other reasons. A combination of these two circuit interrupting devices works as follows. Whenever there is an overload the CB trips whereas for short circuits the HRC fuse operates.

Applications of HRC Fuses

The applications of HRC fuses are enormous but a few very important are: (*i*) protection of cables, (*ii*) protection of bus bars, (*iii*) protection of industrial distribution system, (*iv*) contactor gear for motor control, (*v*) earth faults—both of low and high magnitude, (*vi*) semi-conductor rectifiers and (*vii*) aircraft.

It is to be noted that the HRC fuses cover a very wide range of applications. This involves the principles of fuse design in varying degrees. For special application, the parameters of the fuse are defined to close limits. The design of HRC fuses for the same rupturing capacities for protecting an SCR are different from the one for protecting cables.

An HRC fuse rated for 150 amps continuous rating and 200 kA rupturing capacity at 400 V used for protecting a semi-conductor device weighs about 30 gm whereas an industrial application HRC fuse rated for 100 amps and 250 kA rupturing capacity weighs about 200 gms. The HRC fuses have been used for protecting aircraft equipments and offer many advantages not available by alternative means.

Within wide limits HRC fuses are not affected by frequency. For practical purposes, a fuse tested and rated at 50 Hz is satisfactory for 60 Hz duty and vice versa. As frequency tends towards d.c., the interrupting capability of the fuse at the lower and medium overcurrents may be less, because it is in these zones that d.c. duty is the more onerous. A fuse which has been tested and rated at a given frequency will almost invariably safely interrupt short-circuit faults of higher frequency.

11.8.1 Calculation of fuse rating

In order to select fuse link current rating, the following procedure should be followed :

1. The fuse current rating should be equal to or greater than the design load current for any given section of a wired circuit. From example 11.1, the designed current for light circuit is

$$\frac{3 \times 800}{230} = 10.43 \text{ A}$$

From table 11.3, the appropriate fuse link rating (next higher value) is 15 A and size of fuse is 1.5 sq. mm.

2. In the power circuit the fuse link current rating should be equal to or greater than the value of starting current divided by the factor 2.5. For example (11.3) if we assume that the starting current is about 6 times the full load current; the starting current is

$$21.1 \times 6 \simeq 127 \text{ Amps.}$$

Hence the rating of the fuse link is $\frac{127}{2.5} \simeq 50$ Amp.

From Table 11.4 for the power circuits, the rating of the fuse link (next higher value) comes out to be 60 A. It is to be noted that when severe starting conditions are met with, the starting current is dividing by 1.5 instead of 2.5. These two factors have been obtained through lot of experimentation work.

Table 11.3. Selection of fuse link for lighting load

Current rating of fuse link, a	*Size of conductor (sq. mm)*		
	of wires		
	run exposed	*run in conduit pipe*	*of cables*
10	1.5	1.5	1.5
15	2.5	2.5	1.5
20	4	4	2.5
25	4	4	2.5
35	6	6	6
60	10	10	10
80	16	16	16
100	16	25	25
125	25	35	35
160	35	50	50
200	50	70	70

Table 11.4. Selection of fuse link for Power circuits

Current rating of fuse link a	*Size of conductor (sq. mm.) in wires and cables of branch circuits installed*			*of power mains installed*		
	open	*conduit in pipe*	*in cable runs*	*open*	*in conduit pipe*	*in cable runs*
15	1.5	1	1.5	1.5	1.5	–
20	2.5	1	1.5	2.5	2.5	–
25	4	1.5	1.5	4	2.5	1.5
35	4	2.5	1.5	4	4	2.5
60	6	4	1.5	6	6	4
80	10	4	2.5	10	10	10
100	16	6	4	16	16	16
125	16	10	6	16	16	16
160	25	10	10	25	25	25
200	35	16	16	35	35	35

PROBLEMS

11.1. Name different types of domestic wiring and discuss and compare their performance briefly.

11.2. Explain what specifications for wires used for domestic wiring are normally required.

11.3. Discuss various factors to be considered for deciding size of conductors for domestic wiring.

11.4. Draw a neat diagram of distribution board for domestic wiring and explain the function of each component used.

11.5. Describe different types of cables used for domestic wiring.

11.6. Draw neat-diagram and explain the operation of the following lighting control circuits.

(*i*) Circuit for controlling several lamps with two or more lighting switches.

(*ii*) Control of lamp from two different places and a plug socket.

(*iii*) Control of a group of lamps from several places

(*iv*) Controlling lamps with a chandelier multi-way switch.

(*v*) Controlling a group of lamps from two different-places.

11.7. Discuss the functions of grounding system in distribution networks. What are the effects of bad grounding on personnel working on power system networks ?

11.8. Explain the terms touch potential and step potential, primary and secondary shocks.

11.9. List out some safety measures against electric shocks.

11.10. What are rewirable fuses and HRC fuses ? Discuss their construction and principle of operation and application.

11.11. Explain clearly how a fuse rating is selected for

(*i*) Lighting circuit (*ii*) Power circuit.

11.12. Discuss the advantages of HRC fuse over rewirable fuses.

MULTIPLE CHOICE QUESTIONS

1. In an a.c. circuit having R, L and C in series and operating on lagging $p\,f$, increase in frequency will

(a) reduce the current. (b) increase the current.

(c) (a) and (b) are possible. (d) have no effect on current drawn.

2. In a circuit, voltage and current are given by

$$V = 10 \sin (\omega t + 30°)$$

and $$i = 10 \sin (\omega t - 30°)$$

The power consumed in the circuit is

(a) 100 watts (b) 50 watts

(c) 25 watts (d) 12.5 watts.

3. A series circuit is given by $z = 5 - j12$, its susceptance is

(a) 5/13 (b) 5/169

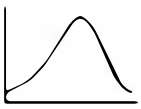
(c) 12/13 (d) 12/169.

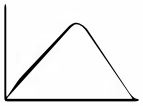
4. In a pure LC circuit (parallel) under resonance condition, current drawn from the supply main is

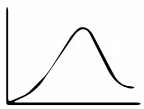
(a) Very large (b) $V\sqrt{LC}$

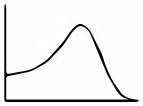
(c) $V/\sqrt{LC}$ (d) Zero.

5. The current response as a function of frequency in a series resonant circuit is given by

(a) 

(b) 

(c) 

(d) 

6. A drawn wire of resistance 5 Ω is further drawn so that its diameter becomes one fifth. Its resistance will now be (volume remaining same).

(a) 625 (b) 125

(c) 25 (d) None of these.

7. A 500 watt 220 V bulb is supplied with 110 volts. Power consumption by the bulb will be

(a) 250 watt (b) 125 watt

(c) slightly more than 125 watt (d) slightly less than 125 watt.

8. In series resonance circuit, increasing inductance to twice its value and reducing capacitance to half its value

(a) will change the resonance frequency.

(b) will change the impedance at resonance frequency.

(*c*) will change the maximum value of current.

(*d*) will increase the selectivity of the circuit.

9. For the circuit shown in figure the value of current I is

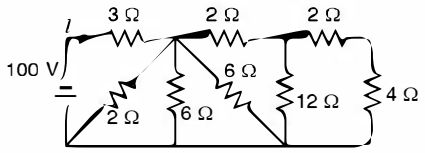


Fig. 1

(*a*) 10 A (*b*) 15 A

(*c*) 20 A (*c*) 25 A.

10. The maximum power that can be distributed in the load in the circuit shown is

(*a*) 3 watts (*b*) 6 watts

(*c*) 6.75 w (*d*) 13.5 watt

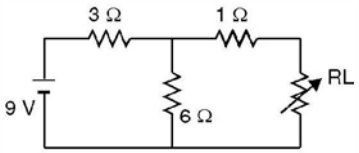


Fig. 2

11. In delta connected circuit when one resistor is open, the power will be

(*a*) zero (*b*) reduced to 1/3

(*c*) reduced by 1/3 (*d*) unchanged.

12. A resistance of 10 ohm is connected in one branch of a network. The current in this branch is 2 A. If this 10 Ω resistor is replaced by a 20 ohms resistance, the current in this branch

(*a*) may be more or less than 2 A (*b*) will be more than 2 A

(*c*) will be less than 2 A (*d*) none of these.

13. If $V = a + jb$ and $I = (c + jd)$, the power is given by

(*a*) $ac - bd$ (*b*) $ac + bd$

(*c*) $bc - ad$ (*d*) $bc + ad$.

14. KCL is a consequences of law of conservation of

(*a*) energy (*b*) charge

(*c*) flux (*d*) all the above .

15. The current in the 1 Ω resistor is

(*a*) 5 A (*b*) 10 A

(*c*) 15 A (*d*) zero.

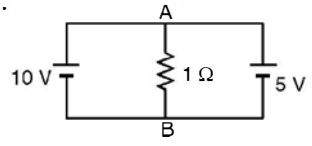


Fig. 3

16. The current in a 5 Ω resistor branch in a linear network is 5A. If this branch is replaced by a resistor of 10 Ω, the current in this branch will be

(*a*) 5 A (*b*) 10 A

(*c*) less than 5 A (*d*) none of the above.

17. The potential of the point A in the given network

(a) 6 V

(b) 7 V

(c) 8 V

(d) none of the above.

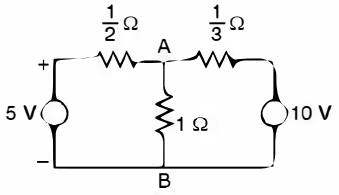


Fig. 4

18. The current through 8 Ω branch is

(a) 1 A

(b) 0.5 A

(c) 1.5 A

(d) none of the above.

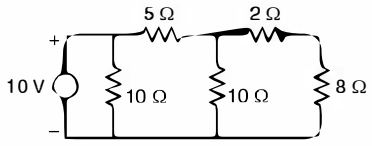


Fig. 5

19. Three identical impedances are connected in delta to a 3-phase supply of 400 V. The line current is 34.65 A and the total power taken is 14.4 kW. The resistance of the load in each phase (in ohms) is

(a) 20 (b) 16

(c) 12 (d) 10.

20. A coil having a resistance of 5 ohms and inductance of 0.1 H is connected in series with a condenser of capacitance 50μ F. A constant alternating voltage of 200 volts is applied to the circuit. The voltage across the coil at resonance is

(a) 200 volts (b) 1788 volts

(c) 1800 volts (d) 2000 volts.

21. If R_g in the circuit shown in the given figure is variable between 20 Ω and 80 Ω, then the maximum power transferred to the load R_L will be

(a) 15 W (b) 13.33 W

(c) 6.67 W (d) 2.4 W.

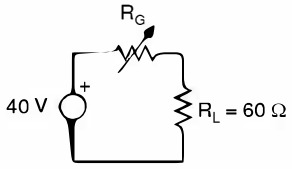


Fig. 6

22. Under the conditions of maximum power transfer from an a.c. source to a variable load

(a) the load impedance must also be inductive, if the generator impedance is inductive

(b) the sum of the source and load impedances is zero

(c) the sum of the source reactance and load reactance is zero

(d) the load impedance has the same phase angle as the generator impedance.

23. In the delta equivalent of the given star-connected impedance Z_{QR} is equal to

(a) 40 Ω (b) $(20 + j10)$ Ω

(c) $\left(5 + j\frac{10}{3}\right)\Omega$ (d) $(10 + j\,30)$ Ω.

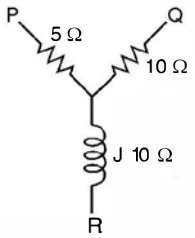


Fig. 7

24. The power delivered to a three-phase load can be measured by the use of 2 wattmeters only when the

(*a*) load is balanced (*b*) load is unbalanced

(*c*) 3-phase load is connected to the source through 3 wires

(*d*) 3-phase load is connected to the source through 4 wires.

25. A 3-phase star-connected symmetrical load consumes P watts of power from a balanced supply. If the same load is connected in delta to the same supply, the power consumption will be

(*a*) P (*b*) $\sqrt{3}P$

(*c*) $3P$ (*d*) not determinable from the given data.

26. Consider the circuit shown in the given figure. For maximum power transfer to the load, the primary to secondary turns ratios must be :

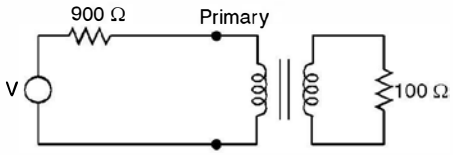


Fig. 8

(*a*) 9 : 1 (*b*) 3 : 1

(*c*) 1 : 3 (*d*) 1 : 9.

27. In two-wattmeter method of power measurement, one of the wattmeters will show negative reading when the load *pf* angle is strictly

(*a*) less than 30° (*b*) less than 60°

(*c*) greater than 30° (*d*) greater than 60°.

28. A 3-phase, 3 wire supply feeds a load consisting of three equal resistors connected in star. If one of the resistors is open circuited, then the percentage reduction in the load will be

(*a*) 75 (*b*) 66.66

(*c*) 50 (*d*) 33.33.

29. An a.c. source of 200 V rms supplies active power of 600 W and reactive power of 800 VAR. The rms current drawn from the source is

(*a*) 10 A (*b*) 5 A

(*c*) 3.75 A (*d*) 2.5 A.

30. Which one of the following networks is the Y equivalent of the circuit shown in Fig. 9 ?

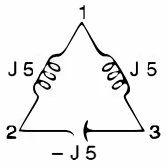
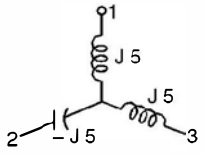
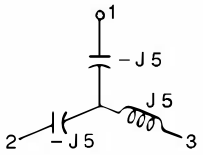
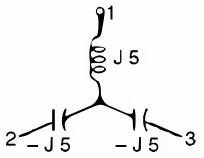
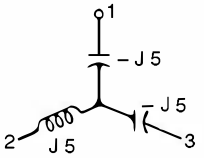


Fig. 9

(*a*) 

(*b*) 

(*c*) 

(*d*) 

31. Which one of the following circuits is the delta equivalent of the star circuit given in Fig. 10 ?

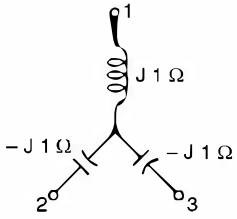
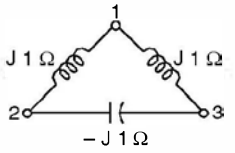
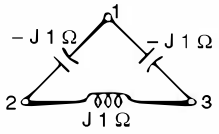
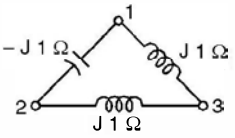
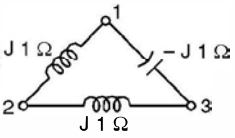


Fig. 10

(*a*) 

(*b*) 

(*c*) 

(*d*) 

32. For the circuit shown in the adjoining figure, the voltage V_{AB} is

(*a*) 6 *V* (*b*) 10 *V*

(*c*) 25 *V* (*d*) 40 *V*.

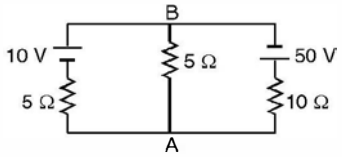


Fig. 11

33. If permanent brake magnet is moved away from the spindle of the moving disc, energy meter

(*a*) will run fast (*b*) will run slow

(*c*) no effect (*d*) will come to a stop.

34. If voltage supply to the energy meter is more than the rated and no load is connected, the energy meter may run

operation can tell the condition of the contacts by looking at the handle position. The MCB is manually resettable after a tripping operation. They are used up to 100A. The trip characteristics are not adjustable. Three are similar breakers called Molded Case Circuit Breakers (MCCB) with current rating up to 3000 Amperes. The trip current of MCCBs may be adjusted in larger ratings.

6.13. Earth Leakage Circuit Breaker (ELCB)

An ELCB detects small leakage currents between live wire and earth and disconnects the equipment. Current operated ELCBs are also called Residual Current Circuit Breaker (RCCB). Voltage-operated ELCBs operate when sufficient voltage appears across the ELCB's sense coil. They have to be manually reset after an operation. They are used when the consumer has a local connection to earth.

6.14. Earthing

An earthing system defines the electrical potential of the conductors relative to the Earth's conductive surface. The choice of the earthing system has implications for the safety and electromagnetic compatibility of the power supply. The regulations for earthing are different in different countries.

A *protective earth* (PE) connection is provided to ensure that all exposed conductive surfaces are at the same electrical potential as the surface of the Earth, to avoid the risk of electrical shock if a person touches a device in which an insulation fault has occurred. This is also called *system earthing*. Earthing of non-current carrying metal works and conductor is essential to the safety of human life, animals, and property is known as *equipment earthing*. Protection against electric shocks has to be provided for direct contact and indirect contact. *Direct Contact* refers to a person coming in contact with a conductor which is live in normal circumstances as shown in Fig. 6.26. Protection against this is called basic protection.

An *indirect contact* refers to a person coming in contact with an exposed-conductive part which is not normally live, but has become live due to an insulation failure or a fault as shown in Fig. 6.27.

The equivalent circuit when the machine is not earthed is shown in Fig. 6.28.

Here R_{body} refers to the resistance of the body the person standing on the ground, R_i the insulation resistance andR_E the resistance of the earth. The current through the body is given by

$V^{Ibody =}$ Ri + Rbody + RE Since R_i is very high, $I_{body} \sim 0$. So under normal conditions no current flows through the body. However, when there is insulation failure, R_i reduces in which case I_{body} is


Earth

Figure 6.26 *Direct contact.*

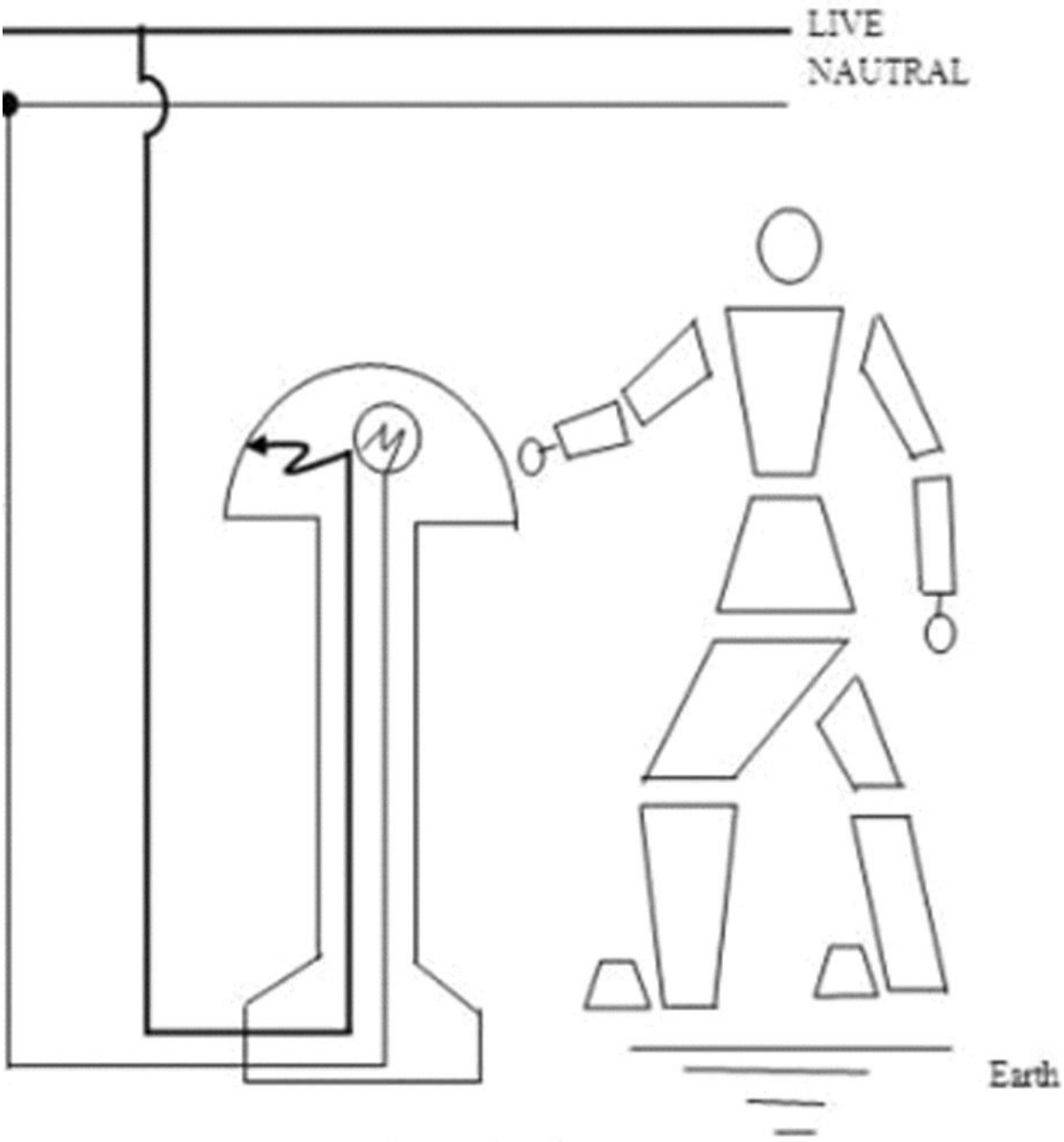


Figure 6.27 *Indirect contact.*

substantial. Now consider when the body is earthed through a wire of resistance R_{E1}, which is around 2–5 . The equivalent circuit under this condition is given in Fig. 6.29. It can be seen that the current now has a parallel path throughR_{E1}. SinceR_{E1} is around 2–5 and R_{body} is of the order 1000–1500 , almost the entire current flows through the

L Machine $I_{body\ RiV\ Rm}$

N Rbody
Earth resistance R_E
Figure 6.28 *No earth.*
L R_i R_m
$N^R E1^R$body
R_E **Figure 6.29** *Body of machine is earthed.*

earthing resistance. The current flowing now through the body is not sufficient to give a shock. In a regular three pin plug, the upper pin is connected to the earth wire in the building. *It is always advisable to use electrical appliances suitably earthed*.

6.14.1. Earthing systems

International Standard IEC 60364 distinguishes three families of earthing arrangement using two-letter codes. The letters are as follows:
(i) The first letter indicates the connection between earth and the power-supply equipment (generator or transformer).
T: Direct connection of a point to earth(from Latin terra for earth). I: No point is connected with earth except through a high impedance (Isolation). (ii) The second letter indicates the connection between earth and the electrical device being supplied.
T: Direct connection of a point to earth
N: direct connection to neutral which is connected to earth.

1) TT system

A TT system has a direct connection to the supply source to earth and a direct connection of the installation metal work to earth as shown in Fig. 6.30.

Generator to transformer R Y B N

Consumer Earth
Figure 6.30 *TT Earthing system.*

TT method is used with overhead lines. In these systems there is no metallic path from the consumer's terminals back to substation transformer secondary windings. Because the earth path resistance may be high a Residual Current Circuit Breaker is often fitted so that if a fault current flows through the earth path a trip disconnects the power supply. The biggest advantage of TT systems is that it is free from high and low frequency noises that come through the neutral wire from various electrical equipment connected to it. That is why TT has

always been preferable for special applications like telecommunications sites that benefit from interference-free earthing.

2) TN system

In a TN system, one of the points in the generator or transformer is connected with earth, usually the star point in a three phase system. The body of the consumer's device is connected to earth via this earth connection. The conductor that connects to the star point in a three-phase system, or that which carries the return current in a *single-phase system* is called *neutral* (N). The conductor that connects the exposed metallic part of the consumer's device is called the *protective earth* (PE). Three variants are possible as shown in Fig. 6.31(a) to (c).

In TN-S system, separate protective earth (PE) and neutral (N) conductors are used from transformer to consumer's device, which are not connected together at any point after the

R R R Y_Y Y B_B B

^{N} PEN^N PE PEN
Consumer Consumer Consumer
(a) TN-S (b) TN-C (c) TN-C-S
Figure 6.31 *TN Earthing system.*

building distribution point. In TN–C system, combined PE and N (PEN) conductor runs all the way from transformer to consumer's device. In TN–C-S, a combined conductor runs up to the consumer's distribution joint, where separate PE and N conductors are used in indoor wiring.

3) IT earthing system
In an IT earthing system, the distribution system has no connection to earth at all as shown in Fig. 6.32.

R Y B

$Consumer_{Earth}$

Figure 6.32 *IT system.*
6.14.2. Methods of earthing

There are different methods for earthing, to ensure low resistance path from appliance to the earth. The various methods of earthing are (i) pipe earthing (ii) rod earthing (iii) horizontal strip earthing and (iv) plate earthing. We will discuss the pipe earthing and plate earthing most commonly used. The extracts are from IS: 3043.

Plate earthing

The earth connection is provided with the help of copper plate or galvanized iron plate of size 60 cm × 60 cm × 6.3 mm for GI or steel. The plate is buried deep in the ground with its face vertical and top not less than 3m below ground. The plate is surrounded by alternate layer of coke and salt for minimum thickness of about 15cm. The earth wire is drawn through a GI pipe and is perfectly bolted to the plate. The earth lead must be G.I. wire or G.I. strip of sufficient cross-section area to carry fault current. A cast iron or mild steel frame with cover having a locking arrangement is embedded in the brick masonary to protect the watering arrangement, which consists of a funnel with a mesh. Salt water is poured periodically through the funnel. The earth electrode should not have a resistance more than three ohms and in rocky soils the resistance may be up to eight ohms. The schematic arrangement is shown in Fig. 6.33.

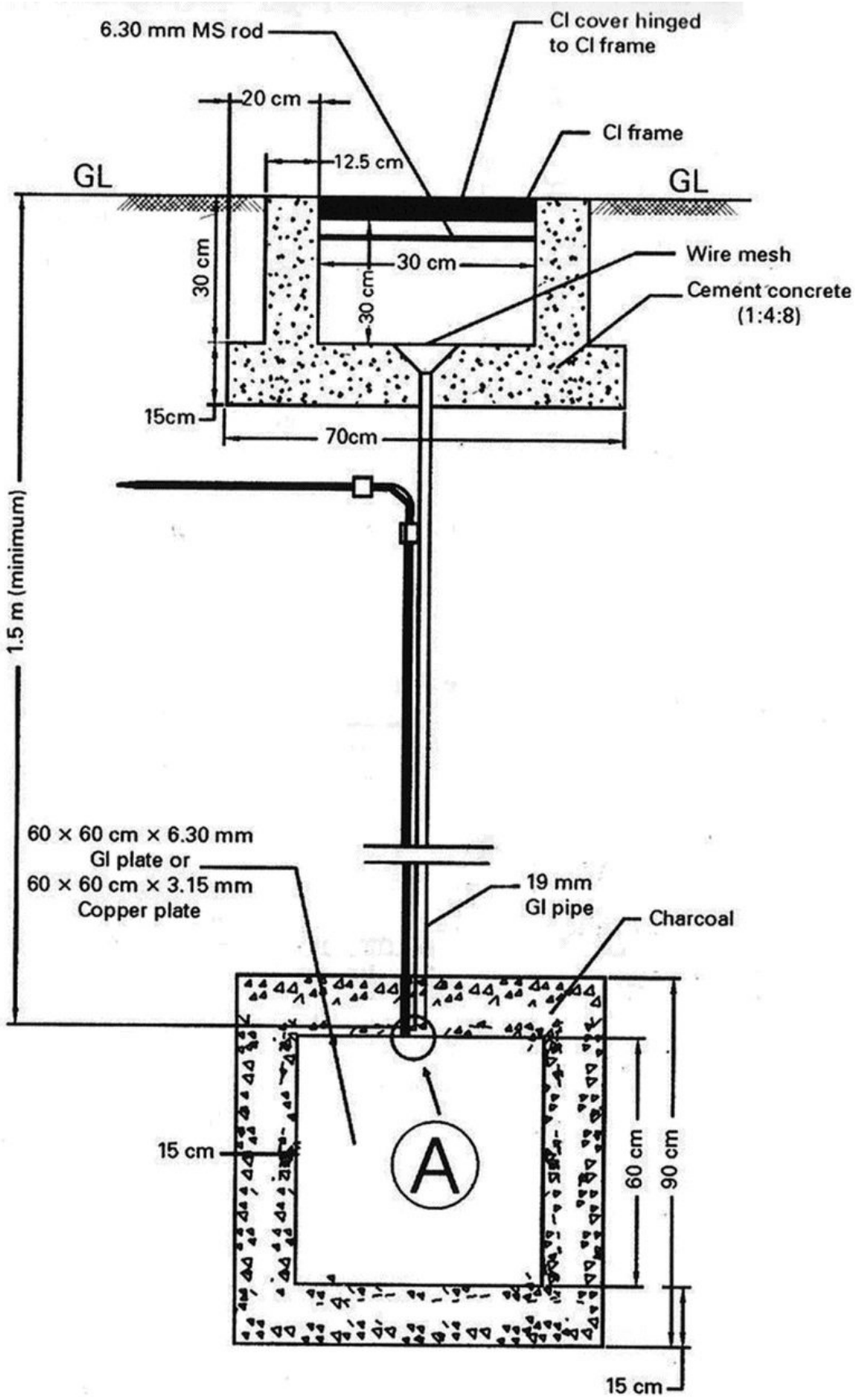

6.30 mm MS rod
CI cover hinged to CI frame
20 cm
CI frame
12.5 cm
GL
GL
30 cm
30 cm
30 cm
Wire mesh
Cement concrete (1:4:8)
15cm
70cm
1.5 m (minimum)
60 × 60 cm × 6.30 mm GI plate or 60 × 60 cm × 3.15 mm Copper plate
19 mm GI pipe
Charcoal
15 cm
A
60 cm
90 cm
15 cm

Figure 6.33 *Plate earthing.*

The main disadvantage is if there is a discontinuity of the earth wire. This cannot be observed physically.

Pipe earthing

In this method a GI pipe of 2 meter length and 38mm diameter is embedded vertically into the ground. This acts as an earth electrode. The depth of the electrode depends on the condition of the soil. The earth wire is fastened to the top section of the pipe through bolts. The pit area around the pipe is filled with salt and coal mixture for improving the conductivity of the soil. Pipe earthing can handle heavier leakage current compared to plate earthing. Water is poured through a tunnel to keep soil wet. The schematic arrangement is shown in Fig. 6.34.

generally rigid, bulky and have large inertia. Hence they are slow and cannot be used to measure rapidly varying quantities. They are also a major cause of noise pollution. Some examples are screw guage, callipers *etc.*

Electrical instruments use electrical principles for measurement. However, most of them have a mechanical moving needle, which limits their application at times. Examples are ammeters, voltmeters, galvanometers *etc.*

Electronic instruments use electronic components for measurement. These instruments respond to quick changes and hence are preferred in modern instruments. Oscilloscopes, digital meters, are some examples of electronic instruments.

7.2.2. Absolute and secondary instruments

Absolute instruments give the magnitude of the quantity to be measured in terms of constants of the instrument. For example the tangent galvanometer, Computation is time consuming, since for every measurement, the quantity has to be computed. *Secondary instruments* are those where the magnitude of the quantity being measured can be read from the output indicated by the instrument. Meters, thermometer, speedometer are a few examples.

7.2.3. Indicating, recording and integrating instruments

Indicating instruments indicate the magnitude of the quantity being measured. A dial, display or a pointer is used to indicate the measurement. Most meters belong to this category.

Recording instruments give a continuous record of the quantity being measured. The commonly used ECG is an example.

Integrating instruments give the integration (summation or accumulation) of the quantity over a period of time. Energy meter, which gives the integration of power consumed over time, is an example.

7.3. Principles of operation

The principle of operation of secondary instruments depends on the effects they utilize. These are briefly discussed.

• *Magnetic effect*: When a current carrying conductor is placed in a magnetic field, it experiences a force. This effect is used in most ammeters, voltmeters,

wattmeters, *etc.*
• *Heating/thermal effect*: The current to be measured is passed through an element which gets heated. The temperature rise can be measured using a thermo cuouple.
• *electrostatic effect*: These instruments use the electrostatic force between two charged plates. Electrostatic voltmeters are an example.

• *Induction effect*: These instruments use the induction effect, wherein an e.m.f. is induced in a disc or drum by induction. It can be used only with a.c. quantities, since induction of e.m.f. requires a changing flux. Induction type wattmeters and energy meters are examples.

• *Hall effect*: If a strip of conducting material carries current in the presence of a transverse magnetic field, an e.m.f. is produced between two edges of the conductor. This is called Hall effect and is used in magnetic measurements and current sensing.

7.4. Essential operating forces in indicating instruments

Indicating instruments indicate the quantity being measured. They have to be calibrated against a standard instrument. Most indicating instruments consist of a needle or pointer which moves over a calibrated scale. The moving system is subjected to three torques:

Deflecting torque or operating torque •
• Controlling torque or restoring torque
• Damping torque.

7.4.1. Deflecting or operating torque

The deflecting torque (T_d) is produced by utilizing one of the effects discussed in Section 7.3. The deflecting torque causes the moving system and hence the pointer attached to it to move from the initial ‘zero’ position, to indicate the value of the quantity being measured.

7.4.2. Controlling or restoring torque

The controlling torque opposes the deflecting torque and increases with the deflection of the pointer. When the two torques are equal, the pointer is brought to rest. The deflecting torque is responsible for producing deflections

proportional to the quantity being measured. If the controlling torque were absent the pointer (being very light) would swing to the maximum position for any magnitude of current. Further, the pointer would not return to zero position on removing the current (or the actuating signal). Controlling torque is also called balancing torque, restoring torque or opposing torque. Its obtained by two common methods

Gravity control

Gravity control is obtained by attaching a small adjustable weight on an arm attached to the moving system. This weight produces a controlling torque by gravity. The arrangement is shown in Fig. 7.1(a).

Assume that the pointer moves by an angle θ, as shown in Fig. 7.1(b). The component of the weight which tries to restore the pointer back to zero position is W sinθ. The controlling torque is thus given by

$T_c = W \sin\theta \times l = Wl \sin\theta = kg \sin\theta$ (7.1) $kg = Wl$ (7.2)

θ l θ

Balance weight

Control

weight

(a) (b)$_W$ W sin θ

Figure 7.1 *Gravity control.*

Thus the controlling torque is proportional to the weight W and the deflecting angleθ.It can be varied by simply adjusting the position of the control weight upon the arm which carries it. The pointer comes to rest when the deflecting torque is equal to controlling torque.

The deflecting torque

$T_d \propto I$ (7.3)

At rest,

$T_d = T_c$ or $I \propto \sin\theta$ (7.4)

Since the current is proportional to sin θ, the scale is non uniform. Instruments using gravity control must be used in vertical position. It is cheap and is not subjected to deterioration with time.

Spring control

The hair spring, usually of phosphor bronze is attached to the moving system as shown in Fig. 7.2.

Balance Control[weight] spring

Volts

0 20 40 60

Pivot

(a) (b) **Figure 7.2** *Spring control.*

When the pointer deflects the spring is twisted in the opposite direction. This produces the restoring torque which is directly proportional to the angle of deflection.

$T_c \propto \theta$ or $T_c = C\theta$; C is spring constant (7.5)

At position of rest,

$T_c = T_d$ or $\theta \propto I$ (7.6)

Since the deflection θ is directional proportional to current, spring controlled instruments have a linear scale. For proper control, the spring should have a fairly large number of turns, to reduce the angular deformation per unit length on full-scale deflection. The springs are made from non-magnetic materials which are not subject to much fatigue. The material of the springs must also have low specific resistance, especially if they are used to lead current in or out of the instrument.

Example 7.1 The torque of an ammeter varies as the square of current through it. A current of 6 A produces a deflection of 90°. What deflection would result for2Aif the instrument is (i) spring controlled (ii) gravity controlled.

Solution: $T_d \propto I^2$ (i) For spring-control, $T_c \propto \theta \therefore \theta \propto I^2$

$90^\circ \propto 6^2$

θ

$^\circ \propto 2^2 \Rightarrow \theta = 90 \times \frac{22}{62} = 10^\circ$

(ii) For gravity-control, $T_c \propto \sin\theta$

$\sin 90^\circ \propto 6^2 \sin\theta^\circ \propto 2^2$

sin

θ

=

sin 90

$\circ \times 2^2\ 1_{62} =_9$

$\sin^{-1}{}^{1}\ 1 = 6.38^{\circ}\theta =_9$

Example 7.2 What would be the deflections if the torque is proportional to the current, in Example 7.1?

Solution: $T_d \propto I$ (i) For spring control

θ

=

90

$\times\ ^2 = 30^{\circ}$ 6 (ii) For gravity control

sin

θ

=

$2_{=}\ 1$

6 3

$\theta = 19.47^{\circ}$

7.4.3. Damping Torque

A damping force acts on the moving system of the instrument only if it is moving. It always opposes its motion. The pointer being made of very light material would vibrate

Under damped Final steady state position

$\text{Deflection}_{\text{Critically}}$ damped Over damped

Figure 7.3 *Damping torque.*

around the final reading. The damping force brings the pointer quickly to rest in the steady position. The various degrees of damping are shown in Fig. 7.3.

The degree of damping should be such that the pointer rises quickly to the steady-state value without overshoot. This is called *critical damping*. If damping is less than this, the system is said to be*under-damped*and the pointer overshoots and oscillates before setting at the steady deflection. If the system is *over-damped* it becomes slow and lethargic.

The common methods of producing damping are (i) air friction (ii) eddy current (iii) fluid friction.

Air friction damping

In one of the arrangements, a light aluminium piston is attached to the moving system as shown in Fig. 7.4(a)

The piston moves in a fixed air chamber closed at one end. The clearance between the piston and the air chamber is small and uniform. Damping is caused by the compression and suction actions of the piston on the air enclosed in the chamber.

In another type of arrangement shown in Fig. 7.4(b), an aluminium vane moves in a sector shaped air chamber. The vane is mounted on the spindle of the moving system. The air suction and compression provides the necessary damping.

Air[Air] chamber[chamber] Vane
Piston Pointer
(a) (b) Spindle
Figure 7.4 *Air friction damping.*
N Damping magnet
Disc Spindle[S] **Figure 7.5** *Eddy current damping.*

Fluid friction is similar to air friction, except that oil is used instead of air. The damping is greater. Oil is generally not preferred because of maintenance problems and the need to maintain instrument in vertical position.

Eddy current dumping

It is also called electromagnetic damping and uses the principle that a conductor moving in a magnetic field has an e.m.f. induced in it. If there is a closed path, this e.m.f. produces a current. A simple form of eddy current damping is shown in Fig. 7.5.

A disc of non-magnetic conducting material like aluminium or copper is mounted on the spindle which carries the moving system and the pointer of the instrument. The disc is placed such that, when it rotates it cuts the magnetic flux between the poles of a permanent magnet. Hence, eddy currents (eddy currents are currents produced on the surface of a conducting material) are produced in the disc. According to Lenz's law, the eddy currents produce a damping force in such a direction as to oppose the very cause producing them. Since the cause here is the rotation of the disc, the eddy current forces, damp the movement of the disc and hence the pointer attached to the spindle.

7.5. Ammeters and voltmeters

Fundamentally there are no diferences in the operating principles of ammeters and voltmeters. They are hence, often classed together. The action of almost all ammeters and voltmeters depends on the deflecting torque produced by a current. In an ammeter the deflection is produced by the current (or a fraction of it) to be measured and in voltmeter the deflection is produced by a current proportional to the voltage to be measured.

Ammeters and voltmeters are connected as shown in Fig. 7.6.

In Fig. 7.6, part of a network where measurements are made are shown. The ammeter is connected in series with the circuit whose current is to be measured. Therefore, it should have a low resistance, so that the voltage across it is small and hence the power absorbed by it.

Similarly, the voltmeter is connected across (in parallel) with the circuit whose voltage is to be measured. Therefore it should have a high resistance, so that it draws very little current and hence power absorbed by it is small. The power lost in an ammeter whose resistance is R_a and through which a current IA flows is I^2R_a Watts. The power lost in a voltmeter whose voltage is V and resistance is R_v is V^2/R_v Watts.

A
$Load_V$ Load

Figure 7.6 *Connection of voltmeter and ammeter.*

7.5.1. Types of ammeters and voltmeters

The main types of ammeters and voltmeters are as follows:

• Permanent magnet moving coil (PMMC). These can be used for only d.c. measurements.
• Moving iron, which uses the magnetic effect of current.
• Electro dynamometer type which can be used for both a.c. and d.c.
• Hot wire; which uses the heating effect produced by current.

• Induction type which uses the principle of electromagnetic induction, and hence can be used only for a.c. measurements.
• Electrostatic type, mostly used for voltmeters.

7.6. Permanent Magnet Moving Coil instruments (PMMC)

These are also called moving coil meters. The operation is based on the principle that when a current carrying conductor is placed in a magnetic field, it is acted upon by a force which tends to move it to one side and out of the field.

7.6.1. Construction

It consists of a permanent magnet and a rectangular coil of many turns wound on a light aluminium former inside which is an iron core as shown in Fig. 7.7.

Pole piece Magnet

Core N S

Figure 7.7 *PMMC construction.*

The U-shaped permanent magnet is made of Alnico and has soft-iron end-pole pieces. Between the magnetic poles is fixed a soft iron cylinder whose function is to make the field radial and uniform and to decrease the reluctance of the air path between the poles and hence increase the magnetic flux.

Surrounding the core is a rectangular coil of many turns wound on a light aluminium frame which is supported by delicate bearings and to which is attached a light pointer. The aluminium frame provides support for the coil and also provides damping by the eddy currents induced in it. The sides of the coil are free to move in the two air gaps between the poles and core. Control is provided by phosphor-bronze hair springs, one above and one below, which additionally serve the purpose of leading current in and out of the coil.

7.6.2. Deflecting torque in PMMC meters

When current is passed through the coil, forces act upon it from both sides producing a deflecting torque. Let

l = length of coil in m

b = breadth of coil in m_{2B} = flux density in Wb/m

N = number of turns of coil.

I = current through coil.

Then force on each side = NIBl Newtons.

Deflecting torque T_d = force× perpendicular distance

= NIBl × b = NIBA N − m.

where A = l × b = surface area of the coil.

For uniform B, T_d is proportional to I.

$T_d \propto I$. When spring-control is used (such instruments are most often spring-

controlled) then the control torque $T_C \propto \theta$. At final steady position, $T_d = T_c$.
$\therefore \theta \propto I$ (7.7) Since the deflecting angle is proportional to current, the scale is linear and uniform. The damping provided by the eddy currents is very effective.

7.6.3. Advantages and disadvantages of PMMC

They have the following advantages

- Low power consumption
- Uniform scale which can be designed to extend over an arc of 170°.
- High torque/weight ratio.
- No hysteresis loss.
- Effective eddy-current damping.
- Not affected by external strong magnetic fields, since they operate in strong fields.

- Can be modified using shunts and resistances to operate over a wide range of currents and voltages.

They have the following disadvantages

- Cannot be used for a.c. When used with a.c., the net deflecting torque would be zero.
- More expensive than moving iron instruments.
- Errors introduced due to aging of permanent magnets and control springs.

7.6.4. Extension of range as an ammeter

For a PMMC meter used as an ammeter, the range can be extended with the help of a low resistance shunt as shown in Fig. 7.8.
The shunt provides a bypass path for extra current because it is connected across the meter. The shunt permits measurement of current many times greater than the normal

Meter $_{I\ Im}$ element

$I_s = I - I_m\ ^{R}m$

R_s (Shunt) **Figure 7.8** *Shunt connection.*

range of the meter. Let

R_s = shunt resistance

R_m = resistance of meter

I_m = Full scale deflection current of meter I = current to be measured

Without the shunt, the maximum current that can be measured is I_m. Referring to Fig. 7.8

$ImRm = Rs(I - Im)$ (7.8)

$R_s = \frac{ImRm}{I - Im} \Rightarrow \frac{I}{Im} = 1 + \frac{Rm}{Rs}$ (7.9)

The multiplying factor is defined as $\frac{I}{Im}$ and is equal to $1+\frac{Rm}{Rs}$. Lower the shunt resistance, higher the multiplying factor. Thus, the shunt increases the current that can be measured by a factor of $1+\frac{Rm}{Rs}$.

The general requirement for shunts are

• Temperature coefficient of shunt and instrument should be low and nearly the same.
• The resistance should not vary with time.
• They should have low thermal e.m.f.

Manganin is generally used for d.c. instruments and constanin is used for a.c. instruments. The meters with external shunts must have leads with a specified resistance to accompany the meter. The leads should not be changed, else serious errors in calibration will occur.

7.6.5. Extension of range as a voltmeter

The PMMC meter can be used as voltmeter with extended range by connecting a high resistance in series with it as shown in Fig. 7.9.

M Vm V

Rse $V - V_m$

Figure 7.9 *PMMC range extension for a voltmeter.*

Let

I_m = full-scale deflection current

R_m = resistance of meter

$V_m = R_m I_m$ = full scale potential drop across

V = voltage to be measured

R_{se} = series resistance connected

The voltage drop across R_{se} is $V - V_m$.

$R_{se} = V - V_m$ or $R_{se} I_m = V - I_m V_m$ V_m

From which

$V = I_m R_{se} + I_m R_{se} = I_m R_{se} + I_m R_m$ (7.10) V_m V_m $I_m R_m$ R_m

The voltage multiplication factor $\frac{V}{V_m} = 1 + \frac{R_{se}}{R_m}$.

Larger the value of R_{se}, greater is the multiplication range. The series resistance is also called *multiplier*. The essential requirements of multipliers are

- Small temperature coefficient
- Resistance should not change with time
- Non-inductively wound for a.c. meters.

Manganin and constanin are two materials commonly used. Multipliers are mounted inside the instrument case for voltages upto 50 V. For higher voltages they may be mounted separately.

Example 7.3 A moving coil ammeter has a fixed shunt of 0.015 with a coil circuit resistance of 750 . A voltage of 0.5 V produces full-scale deflection in the meter. To what total current does it correspond? What is the value of shunt needed to give a full scale deflection when the total current is 50 A?

Solution: 750Ω

0.5V

$I_m = \frac{0.5}{750} = 6.66 \times 10^{-4}$ A

$\frac{0.5}{0.015} = 33.33$ A $I_s =$

Total current is 33.33399 A.

When total current is 50 A for full scale deflection,

$I_s = 50 - 6.66 \times 10^{-4} = 49.999334$ A

$I_m R_m = 6.66 \times 10^{-4} \times 750 = 9.99 \times 10^{-3} R_s = I_s$ 49.999334

Example 7.4 A moving-coil instrument has a resistance of 10 and gives full scale deflection when current through it is 50 mA. How can the range be extended to measure (i) 500 A (ii) 1000 V.

Solution: (i)

500A 10Ω

50mA

499.95A

$R_s = I_m R_m = 50\times 10^{-3} \times 10 = 0.001 I_s 499.95$
(ii)
0.05A
10Ω 1000V
R_{se}
$V = 1 + R_{se}$; $V_m = 0.05\times 10 = 0.5$ $V_{Vm\ Rm}$

$1000_{1+}\ R_{se}$
$0.5 = 10$
$R_{se} = 19.99$ K.

Example 7.5 A galvanometer has the following parameters:
B = 10 mWb/m^2; N = 250 turns,l × b = 15× 15 mm^2 K = 12× 10^{-9}N − m/radian. What is the deflection of the galvanometer when a current of 1 mA flows through it.
Solution:
Td = NBIA = (250) × (10× 10^{-3}) × (1× 10^{-6}) × (225× 10^{-6}) = 562× 10^{-12}N − m
$T_c = K \times \theta = 12\times 10^{-9} \times \theta$
At steady state $T_d = T_c$,
$12\times 10^{-9} \times \theta = 562\times 10^{-12}$ θ = 0.04683 rad = 2.68°

7.7. Moving iron instruments

Moving iron instruments are of two types: attraction type and repulsion type. In both types, the principle of operation is the force experienced by an iron piece in the presence of a magnetic field. The necessary magnetic field is produced by the ampere-turns of a current-carrying coil. When used as an ammeter the coil has fewer turns of thick wire so that the ammeter has low resistance. When used as a voltmeter, the coil has high impedance so as to draw as small current as possible since it is connected in parallel. Since current through it would be small, it has to have a large number of turns to produce the necessary ampere turns.

7.7.1. Attraction type instruments

The attraction type instrument is shown in Fig. 7.10. It uses the principle that a soft iron brought near either end of a current carrying coil gets attracted into the coil.

In the most common type, an oval shaped disc of soft iron is pivoted on a spindle between bearings near the coil. The iron disc swings into the coil when a current

between bearings near the coil. The iron disc swings into the coil when a current is passed through the coil. The field strength is strongest at the centre of the coil. Therefore, the iron disc is pivoted in such a way that the bulk of the iron moves into the centre of the coil.

A pointer attached to the spindle carrying the disc, would deflect when current flows through the coil. The deflection is proportional to the magnetic field which is proportional to the current producing it. Whatever is the direction of the current, the iron disc would always be magnetized in such a way that it is pulled inwards. Therefore, these instruments can be used for both a.c. and d.c. currents.

Pointer C Air chamber
Balance weight Moving iron Control weight Coil winding C **Figure 7.10** *Attraction type instrument.*
φ
θ
H
Figure 7.11 *Deflecting torque.*

7.7.2. Deflecting torque in attraction type MI instruments

Let the axis of the disc make an angle φ with a direction perpendicular to the direction of the fieldH produced by the coil, when in zero position. Let the angle beθ, corresponding to a current I through the coil.

Component of H along axis of disc is proportional toH cos[90−(φ+θ)]orH sin(θ +φ). The force is proportional to H^2 sin(θ + φ). When permeability is constant, then H ∝ I. Therefore, F ∝ I^2 sin(θ + φ). Assuming the force acts at a distance l from the pivot of the rotating disc, T_d = Fl cos(θ + φ).

∴ T_d ∝ I^2 sin(θ + φ) × l cos(θ + φ) (7.11) ∝ I^2 sin 2(θ + φ) = KI^2 sin 2(θ + φ) (7.12) Assuming spring-control, T_c ∝ θ.

Section of coil
Coil
Magnetic field of Fixed coil iron$_A$
B Moving
iron
Air
chamber Pivot Iron rods lower one fixed **Figure 7.12** *Repulsion type MI instrument.*

At steady state, $KI^2 \sin 2(\theta + \varphi) = K\theta$ or $\theta \propto I^2$. When a.c. is used, $\theta \propto I^2_{.rms}$
With gravity control, $\sin\theta \propto I^2 \sin 2(\theta + \varphi)$.

Direction of force
With either of the control, the scale is non-uniform. Damping in this arrangement, is provided by air friction damping.

7.7.3. Repulsion type MI instruments

The principle used is that like magnetic poles repel. The construction is shown in Fig. 7.12.

Two soft-iron rods, are placed parallel to one another and along the axis of a coil which is fixed. One of the soft-iron pieces is fixed and the other is movable and carries the pointer which moves over a calibrated scale. The current to be measured is passed through the coil. It sets up a magnetic field which magnetizes the two rods similarly. This causes a force of repulsion between the two rods which results in the pointer being deflected. The force of repulsion is proportional to the square of the current passing through the coil. Irrespective of the direction of current, a force of repulsion exists since the rods are always magnetized similarly.

Different shapes of iron pieces are used instead of rods, to achieve a uniform scale. The deflecting torque is proportional to I^2. For a.c. currents it is proportional to $I^2_{.rms}$ The instruments are generally spring-controlled. Eddy current damping cannot be used in these instruments since, the presence of a permanent magnet would affect the deflection. In terms of inductance, the deflecting torque is given by $T_d = {}^1I^{2dL}$ N-m, where L is

the self inductance of the coil.$^{2\ d\theta}$

7.7.4. Advantages and disadvantages of MI instruments

Advantages are

- They are cheap
- They are reliable
- They can be used for both a.c. and d.c.

They have following disadvantages.
- Calibration is not accurate with d.c.

• Due to hysteresis in the iron parts of the moving system, the readings are higher for descending values but lower for ascending values.
• Stray magnetic fields affect the reading.

• With a.c. currents the reading depends on the frequency, since frequency affects the impedance of the coil and the magnitude of the eddy currents. Therefore, an instrument calibrated at a particular frequency would be erroneous at another frequency.

7.7.5. Extension of range

The range can be extended to extend the range as an ammeter or voltmeter, just like PMMC meters. With d.c. currents there is no problem, but with a.c. currents the frequency is important. Consider the connection of shunt as shown in Fig. 7.8. Now,

Current through meter $= I_m = R_s + jwL_s = Z_s$ .Current through shunt Is Rm + jwLm Zm

Where R_s,L_s are resistance and inductance of the shunt and R_m,Lm are resistance and inductance of instrument. The current is distributed in the inverse ratio of their impedances. If the ratios,

$\frac{Lm}{} = \frac{Ls}{}$, then the multiplying factor$_{Rm\ Rs}$

$I = 1 + \frac{Rm}{}$

$I_m\ R_s$

When a series resistor is connected to extend range as voltmeter, then the voltage magnification

$V_{= 1+}$ Rse Vm Rm When used with a.c. it is essential that the total impedance of the voltmeter and the series resistance remains constant over the different frequencies it is used.

Example 7.6 A 250 V moving iron voltmeter takes a current of 0.05 A when connected to d.c. If the coil inductance is 1.2 H, determine the reading when connected to a 250 V, 100 Hz supply.

Solution: When used with d.c., the meter offers only resistance= $\frac{250}{0.05}$ = 5000 . When used with a.c., the impedance is given by

$Z = R^2 + (L\omega)^2 = 5000^2 + (1.2\times 100\times 2\pi)^2$ 5056.53=

voltage of the instrument = $250 \times \frac{5000}{5056.53} = 247.2$ V

7.8. Errors in meters

7.8.1. Errors in PMMC meters

The main source of error in moving coil instrument are due to

(i) Weakening of springs due to ageing and temperature effects.
(ii) Change of resistance of the moving coil with temperature
(iii) Weakening of permanent magnets due to ageing and temperature effects.

The effect of weakening of springs on the performance of the meter is opposite to that of ageing of magnets. The weakening of magnets tends to decrease the deflection for a particular value of current, while weakening of springs tends to increase the deflection. Normally, 1°C rise in temperature reduces strength of springs by 0.04% and reduces flux density in the air gap by 0.02%. Hence, the net effect is to increase the deflection by 0.02% per °C rise.

The moving coil is wound with copper wire having a temperature coefficient of 0.004°C. The deflection would reduce by about 0.04% per °C rise in temperature when used as a milli ammater. When used as voltmeter, since a large series resistance is connected, this error is negligible.

When shunts are connected to extend the range as an ammater, errors are introduced because of the difference in the temperature coefficients of the copper coil and the manganin shunt. To reduce the error a small resistance called swamping resistance, made of manganin, is connected in series with the coil. It is also used as an adjustment for final calibration.

7.8.2. Errors in MI meters

Hysteresis error occurs with both a.c. and d.c. This is due to the fact that flux density is different for the same value of current, when current is increasing and when it is decreasing. The flux density is higher while current is descending. The reading therefore tends to be higher for descending values of current than for ascending values. This error can be minimized by making the iron parts small so that they demagnetize quickly. Hysteresis can produce around 2–3% error. Nickel iron alloys have narrow hysteresis and can bring down error.

Errors due to temperature are mainly due to the springs weakening with temperature rise. Stray magnetic fields distort the operating magnetic field

temperature rise. Stray magnetic fields distort the operating magnetic field, which is relatively weak.

Some errors occur only with a.c. quantities. The primary source of this error is the variation in the impedance of the working coil and change in the magnitude of eddy currents set up in the metal parts of instruments. At low frequency the eddy current error increases with square of the frequency while at high frequencies the error is practically constant around 0.1%.

7.9. Electrodynamometer Wattmeter

The Wattmeter has two coils - the *current coil* and the *voltage coil* (also called pressure coil). The current coil is connected in series with the load and carries the load current, while the potential coil is connected across the load and therefore carries a current proportional to the voltage. A high non-inductive resistance is connected in series with the pressure coil to limit the current through it.

7.9.1. Construction

(1) **Fixed coils:** The current coil forms the fixed coil and carries the load current. Current coils are made fixed because they can then be constructed to carry heavy currents. The current range is restricted to 20A, to avoid eddy currents.If higher currents are involved, it is better to use the wattmeter in conjunction with a CT. The current coil is wound in two halves positioned parallel to each other and distance adjusted to give uniform magnetic field.

(2) **Moving coil:** The pressure coil is the moving coil and it is mounted on a pivoted spindle. Spring control is used for the movement. The current through this is limited to 50mA.

(3) **Control:** Spring control is used.
(4) **Damping:** Air friction damping is used. The dynamometer wattmeter connections and constructions are shown in Fig. 7.13.
I
2

M
Current coil

FF_{I1}
I1 $I1_I$ I_2
[1] $Supply\ Voltage_R$ Load High
coil
Supply V
R

Figure 7.13

7.9.2. Deflecting torque

The flux density is directly proportional to the currentI_1 (current through the current coil), since the coils are air cored.

$B \propto I1$ or $B = K1I1$
$I2 \propto V$ or $I2 = K2V$
$Td \propto BI2 = KV\ I1 = K \times power$

For *dc*, the power is given by product of voltage and current. Let us see what happens in case of *ac*. Letv = instantaneous value of voltage across moving coil.i = instantaneous current through fixed coil. $T_{d(inst)} \propto vi = Kvi$.

However, because of inertia of the moving system, the instrument indicates the mean or average power.
Therefore mean deflecting torque $T_m \propto$ average value of v_i. Let
$v = V_m \sin\theta$ and $i = I_m \sin(\theta - \varphi)$ 1
2π
$\therefore T_m \propto_{2\pi\ 0} V_m \sin\theta \times I_m \sin(\theta - \varphi)d\theta$
$\propto$
$V_m I_m\ ^{2\pi}$
2
π
sin
$_0\ \theta \sin(\theta - \varphi)d\theta$
$\propto V_m I_m\ ^{2\pi} \cos\varphi - \cos(2\theta - \varphi)_{d\theta 2\pi\ 0\ 2}$
V
m
I
m

θ
cos
φ
−
sin
(
2
θ
−
φ)
2π
∝ 4π 2_0
∝ $^{Vm\ I\ m}$2 cosφ ∝ VI cosφ$_{\sqrt{2}}$√
Therefore T_m ∝ VI cosφ which is the active power.

Thus the deflection is proportional to the active power. The scales of dynamometer wattmeter are uniform because deflection is proportional to average power and for spring control, the controlling torque is proportional to deflection.

(i) **Error due to wattmeter connections:**
There are two ways in which a wattmeter can be connected as shown in Fig. 7.14.

In commercial wattmeters the current coil is marked M(mains) andL(Load) and the voltage coil marked C − V. In Fig. 7.14(a), the current through the current coil is I_L, where as the voltage across the voltage coil is the phasor sum ofV_L andV , drop across current coil. The phasor diagram is shown in Fig. 7.15(a).

ML IL ML IL
$^{C\ V}$ V_L Load$^{C\ V}$ V_L Load
(a) (b) **Figure 7.14** *Wattmeter connection.*
^{VL} V' $_{\varphi\ \theta\ V\ \varphi\ \theta\ I}$ V = V_L
$^{IL\ I'}$(a) I_L = I (b)
Figure 7.15
The Wattmeter reading = VI_L cosθ whereV = V_L+ V andθ is phase difference between V and I_L.

$VI_L \cos\theta = (V_L \cos\varphi + V)I_L$
$= V_L I_L \cos\varphi + V I_L$
= power in load+ power in current coil.

The second connection is as shown in Fig. 7.15(b).
Here the current coil carries the current of the voltage coil and the load current while the voltage across the voltage coil isV_L.

$W = V_L I \cos\theta = V_L(I_L \cos\varphi + I) = V_L I_L \cos\varphi + V_L I$ = power in load+ power in pressure coil.
Thus, there is always error in the Wattmeter reading.
(ii) **Error due to inductance of voltage coil:**
The error in wattmeter reads when the inductance of pressure coil is considered is given by,
$\sin\alpha$ % error $= \cot\varphi + \sin\alpha \times 100$
where$\alpha = \tan^{-1} XP$ whereX_P is reactance of pressure coil andR_P is resistance of$_{RP}$pressure coil circuit (includes the high resistance connected in series).
(iii) **Error due to stray fields:**
The operating field of such an instrument is very small. Hence, it is liable to stray field errors.
(iv) **Error due to eddy currents:**

The eddy current produced in the solid metallic parts of the instrument by the ac field of the current coil, changes the magnitude and strength of the operating field thus producing an error in the reading of the wattmeter.

(v) **Error due to capacitance of voltage - coil circuit:**

There is always some small amount of capacitance in the voltage - coil circuit, particularly in the series resistor. This actually reduces error due to the inductance of the voltage coil.

Example 7.7 A 250V, 10A dynamometer Wattmeter has resistance of current coil of 0.5 and that of potential coil of 12,000 . Find the error due to the two connections of Fig. 7.14, if used to measure power of a 250V upf load drawing 10A.

Solution:

(i) Consider connection of Fig. 7.14(a).
Load power = 250× 10 = 2500W Power in current coil = $10^2 \times 0.5 = 50$W
Wattmeter reading = 2550w
% error$^{50} \times 100 = 1.96\% =_{2550}$
(ii) Consider connection of Fig 7.14(b)
% error Load power = 2500W 250
2
Power in voltage coil $=_{12000} = 5.2$W
$^{5.2} \times 100 = 0.207\% =_{2505.2}$

7.10. Energy meter

Induction type energy meters are the most commonly used. They measure electric energy in kilowatt hours (KWh).

7.10.1. Construction

The energy meter is shown in Fig. 7.16. There are four main parts.
C_2 M_2
Supply ^{V}voltage$_{\Phi 2\ C}$ D $^{\Phi 1\ C1}$ M_1 II (a)
S N
N S
Braking magnet (b)
Figure 7.16

(i) **The driving system:** The driving system of the meter consists of two electromagnets, the core of them made of silicon steel laminations. The electromagnet M_1 is called the *series magnet* and is excited by the line current. The flux φ_1 produced by it is in phase with the line current and proportional to it. The second magnet M_2 called the *shunt magnet* is connected across the supply voltage and therefore the coil carries a current proportional to the supply voltage. *Copper shading bands* C are provided on the center limb of the shunt magnet to bring the flux φ_2, produced by the shunt magnet exactly in quadrature with the line voltage. The position of these bands is adjustable.

(ii) **The moving system:** This consists of an aluminium disc mounted on a light alloy shaft, positioned in the air gap between the two magnets. The disc cuts the fluxes of both the magnets.

(iii) **Braking system:** A permanent magnet positioned near the edge of the disc forms the braking system. The disc moves in the field of this magnet and thus

provides a braking torque. The position of the braking magnet is adjustable and hence the braking torque can be adjusted by shifting the permanent magnet to different radial positions.

(iv) **Resistering system:** It is the mechanism used to record continuously a number which is proportional to the revolutions made by the moving system as shown in Fig 7.17.

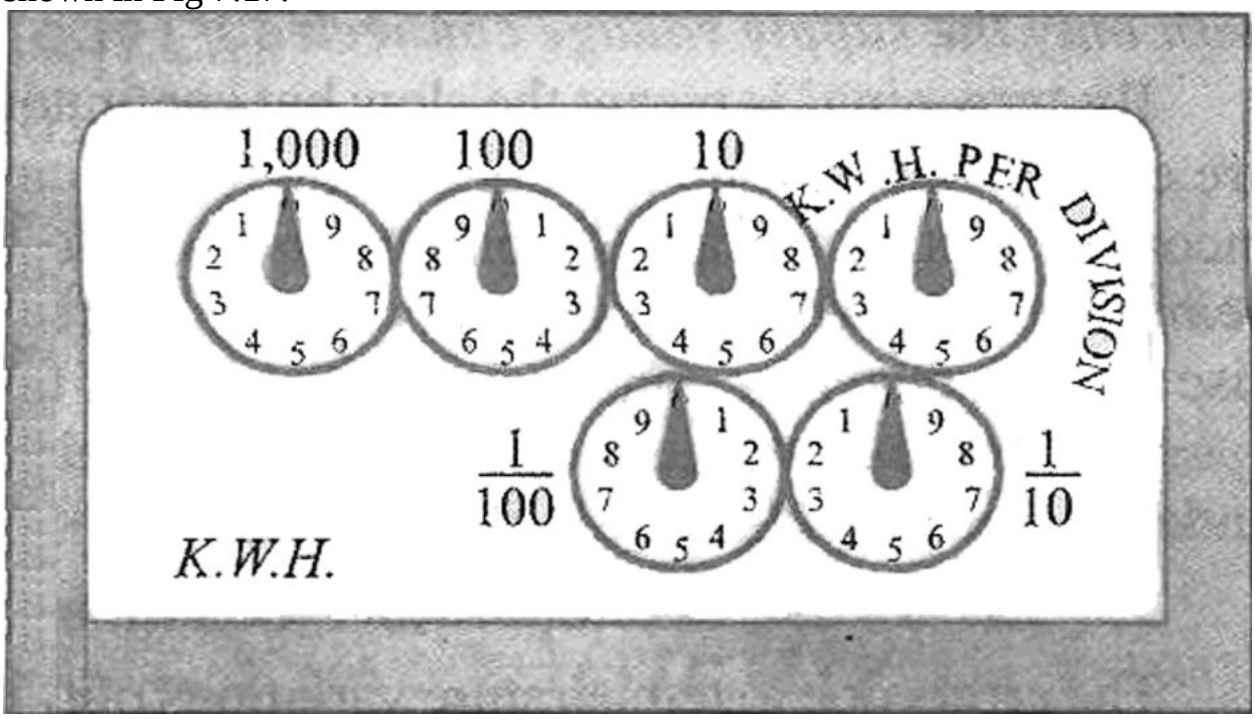


Figure 7.17

7.10.2. Deflecting torque

Consider the phasor diagram shown in Fig. 7.18.

$\varphi_1 \propto I$ and in phase with it

$\varphi_2 \propto {}^{V}$ and in quadrature with $it_{L\omega}$

V I

$\varphi^{\varphi 1}$ α φ_2 **Figure 7.18**

$T_d = K\omega\varphi 1_m \varphi 2_m \sin\alpha$

$\omega I\ V \sin(90^- \varphi) \propto VI \cos\varphi \propto$ power$= K2$

ωL

also

$T_b \propto$ N. (braking torque)

The disc achieves a steady speed N when the two torques are equal.

$T_d = T_b$

$\therefore$ N ∝ power W

In a given period of time, the total number of revolution^{t}Ndt is proportional to^{t}Wdt, the electric energy consumed.$^{0\,0}$

7.10.3. Errors in energy meters

(i) **Phase and speed errors:** The flux due to the shunt magnet does not lag behind the supply voltage by exactly 90° because of the resistance of the pressure coil. This causes error, which can be compensated by means of the copper shading band. These bands are also called *power factor compensator*. The error in speed of the meter can be adjusted by adjusting the position of the brake magnet.

(ii) **Creep:** In some meters a slow but continuous rotation is obtained even when there is no current flowing through the current coil and only the pressure coil is energized. This is called *creeping*. The major reason for this is over compensation of friction. Frictional forces at the rotor bearings and in the resistering mechanism give unwanted braking torque on the disc. To compensate for this, it is necessary to provide for a small torque in the same direction as the deflecting torque and as nearly equal as possible to the frictional torque. This is obtained by means of a small shading loop situated between the center pole of the shunt magnet and the disc and slightly to one side of the center-line of the pole. The interaction between the portion of the flux which are shaded and unshaded by this loop and the currents they induce in the disc, results in a small driving torque whose value can be adjusted by lateral movement of the loop.

An over compensation produces a driving torque even when current is zero, giving rise to creeping. To prevent creeping, two holes are drilled on the disc, which are diametrically opposite. The disc comes to rest when one of the holes comes under the edge of a pole of the shunt magnet.

(iii) Errors are produced because of temperature variations. These are in general negligible.

Questions

1 What are direct and indirect methods of measuring quantities?
2 How are instruments classified?
3 What are indicating instruments?
4 What are the various effects used in production of deflecting torque?

5 Explain how controlling torque is produced using gravity control. What are its disadvantages?

disadvantages?

6 What is damping torque? How is it produced?
7 What are the various types of ammeters and voltmeters?
8 Clearly explain the constructional features of a permanent magnet moving coil meter.
9 Why can't PMMC meters be used for a.c.?
10 Show that in a PMMC meter, the scale is linear.
11 What are the advantages and disadvantages of PMMC instruments? 12 Describe the working principle of an attraction type MI meter.

13 How can an MC meter be used (i) as an ammeter with extended range (ii) an voltmeter with extended range?
14 What are sources of errors in meters?
15 What is a repulsion type MI meter?

16 A PMMC meter has coil dimensions 15×12 mm^2. The flux density in the air gap is 1.8×10^{-3} Wb/m^2 and the spring constant is 0.14×10^{-6} M-m/rad. Determine the number of turns required to produce an angular deflection of 90° when a current 5 mA flows through the coil.
Ans (N = 136)

17 A 1 mA meter with an internal resistance of 100 is to be converted into a 0–100 mA ammeter. Calculate the shunt resistance required.
Ans 1.0 .

18 Find the multiplying power of a shunt of 200 resistance used with a galvanometer of 1000 resistance. Determine the value of the shunt resistance to give a multiplying power of 50.
Ans (i) 6 (ii) 20.4 .

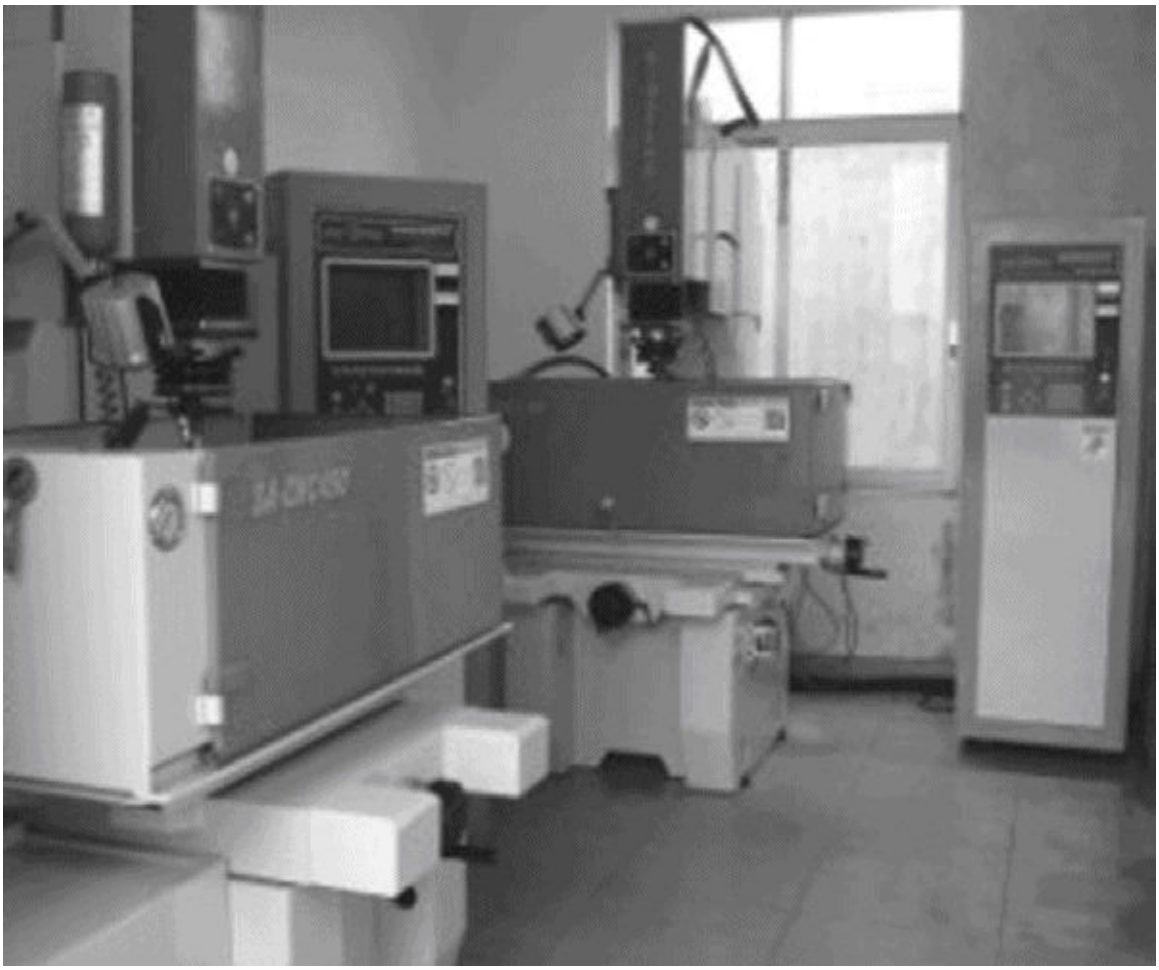
19 A moving coil instrument gives a full scale deflection of 10 mA when the potential difference across its terminals is 100 mv. Calculate (i) the shunt resistance for a full scale deflection of 100 A (ii) the series resistance for a full scale deflection of 1000 V. What is the power dissipated in each case.
Ans (i) 0.001 ; 10 W (ii) 100 K; 10 W.

20 What are the advantages and disadvantages of moving iron instruments?

Part II
Electric Machines

CHAPTER - 1

Introduction to Electric Machines



Learning objectives

Electric energy
Basic generator and motor Basic principles of operation Types of machines
• d.c. machines
• synchronous machines
• induction machines
• transformers
Construction of machines Losses in machines

1.1. Electric Energy

Electric energy has invaded our lives in a dramatic manner, ever since Edison invented the first electric bulb. Electric appliances like fans, washing machines, mixers, blenders, grinders, ovens *etc.* have found widespread use. The main advantage of electric energy over other forms of energy is that it can be transmitted over long distances easily with high efficiency.

Electric machines are the most important energy converters. Electric generators convert mechanical energy to electric energy. The source of the mechanical power is called prime mover. The conversion process is shown in Fig. 1.1.

An electric motor converts electric energy into mechanical energy as shown in Fig. 1.2. A transformer is a static device that transforms electric energy from one a.c. voltage level to another. This device is responsible for making the electric system predominantly a.c.

$P_{electrical}$

T_e

Primer Electric mover generator T_M : Mechanical torque

T_e : Electrical torque

T_M ω

P

Shaft

$mechanical_{Losses}$

Figure 1.1 *Electric generator.*

$P_{electrical}$

T_L

Electric Mechanical motor load

T_M ω

Losses Pmechanical

Figure 1.2 *Electric motor.*

1.2. Basic principles of electric machines

The electromechanical conversion is based on two principles:

(i) A current carrying conductor placed in a magnetic field experiences a force. The force is given by

The force is given by
$f = li \times B$

where
l = Length of conductor
i = Current
B = Magnetic field strength
f = Force.

Motors make use of this principle.
(ii) When a moving conductor is placed in a magnetic field, an e.m.f. is induced in it, given by
$e = lv \times B$

where
e = e.m.f. induced
l = Length of conductor v = Velocity of conductor B = Magnetic field strength.

1.3. Basic types and constructional features The basic rotating electric machines are

• The d.c. machines
• The polyphase induction machine (a.c.)
• The polyphase synchronous machine (a.c.)
• Single phase induction machine (a.c.) All rotating machines consist of two main parts:

(a) Cylindrical rotating member called the rotor and a stationary member called stator, with air gap in between the two. The rotor has an axial shaft which is carried on bearings at each end located in end covers bolted to the stator. The shaft extends out of the end cover usually at one end and is coupled to either the prime movers or the load.

(b) The stator and rotor are both made of a magnetic material, mostly steel, which conducts the magnetic flux upon which the electromechanical energy conversion depends. In d.c. machines and synchronous machines, the main field is created by the field poles excited by d.c. current. The winding on the field poles is called the field winding. The other winding is called the armature winding.

1.3.1. D.C. machines

In d.c. machines, the armature is the rotor and the field is the stator. The field poles are even in number and are symmetrically placed around the armature. As the armature rotates, alternating e.m.f. and current are induced in the armature. It is rectified to d.c. using a commutator. It is cylindrical in shape and consists of wedge shaped copper segments bound together while they are insulated from each other. The armature is made of laminated steel with slots cut out on the periphery to house the armature winding. The ends of each armature coil are connected to the commutator segments to form a closed winding.

1.3.2. Synchronous machines

In synchronous machines the rotor carries the field poles, on which the field winding is wound. The field poles are excited by direct current. The armature which carries a three phase winding, wound for the same number of poles as the rotor, is the stator. When the rotor rotates, it produces an alternating e.m.f. in each of the armature windings, forming three phase balanced voltages of frequency

$$f = \frac{NP}{120},$$

where
f = Frequency
N = Speed of rotor in r.p.m.
P = Number of poles.

1.3.3. Induction machines

Induction motors are the most rugged machines. The stator winding is a three phase winding, similar to the armature winding of the synchronous machine, when three phase voltage is applied to the stator winding, a synchronously rotating field is produced. The rotors are of two types:

Squirrel cage rotor: In this, the rotor has copper or aluminium bars embedded in slots which are short circuited at each end. It is an extremely rugged construction, but develops a low starting torque.

Slip-ring (or wound-rotor) rotor: The rotor has proper 3-phase winding with

three leads brought out through slip-rings and brushes. The leads are normally short circuited when the motor is running. Resistances can be introduced in the rotor circuit, via the slip-rings, to improve the starting torque.

In induction motors, the rotating field created by the stator winding moves past the rotor conductors, inducing currents in them. These induced currents produce their own field. Torque is developed by the interaction of the stator field and the field produced by the rotor currents. The torque rotates the rotor at a speed slightly less than the synchronous speed.

1.3.4. Single phase a.c. motors

These machines are generally induction type machines. Single-phase a.c. motors are employed for low-voltage, low-power applications. They are fractional kW motors. They have reduced motor torque and the torque is pulsating.

1.3.5. Transformers

A transformer is a static device comprising coils coupled magnetically. The transformer is used (i) to change voltage and current levels in circuits (ii) match source and load impedances for maximum power transfer (iii) electrical isolation. The transformer consists of two insulated windings, interlinked by a common or mutual magnetic field established in the core of the magnetic material. One of the windings called the primary winding is connected to an alternating voltage source, setting up an alternating flux in the core. This mutual flux links with the second winding called the secondary winding. A voltage is induced in the secondary winding, at the same frequency as the supply voltage. The magnitude depends on the number of turns. The ratio of the primary voltage to the secondary voltage is equal to the ratio of their respective turns. If the secondary voltage is greater than the primary, the transformer is called step-up. If it is less it is called step-down.

1.4. Losses in Machines

Losses occur in all types of machines, resulting in their efficiency being less than 100%. The efficiency of a machine is defined as

$$\% \eta = \frac{\text{Output}}{\text{Input}} \times 100\%$$

Output and input are measured in watts. The losses in machines are as follows.

1.4.1. Copper losses

Copper losses occur in all windings and account for the heat lost in the resistance of the winding. If the resistance of the winding is R, and it carries a currentIA, the copper loss is equal to I^2R watts.

1.4.2. Iron losses

Iron losses occur in the magnetic material of the machines. Iron losses are due to eddy currents and hysteresis.

Eddy current losses: These occur because of currents induced on the surface of the magnetic materials. These losses can be reduced by using laminations.

Hysteresis losses: These losses occur due to magnetic reversals.

1.4.3. Mechanical losses

These losses are associated with moving parts friction and windage losses form a major portion of the mechanical losses.

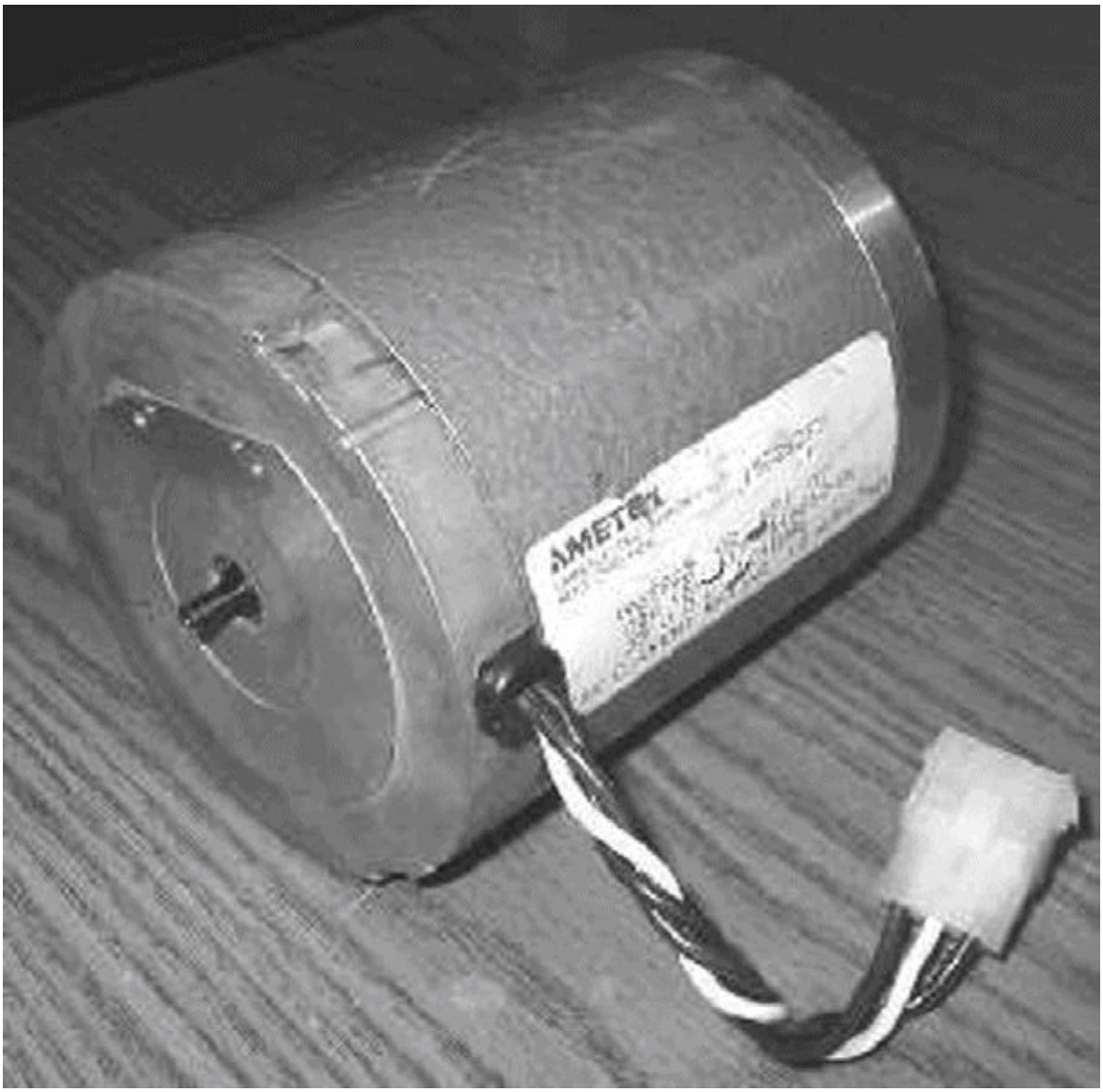
In the subsequent chapters we deal with each of the machines in detail.

Questions

1. What is a generator?
2. What is a motor?
3. Explain the different machines available.
4. What is the principle of operation of a generator?
5. What is the principle of operation of a motor?
6. Explain the working of a transformer.
7. What are the various losses occurring in electric machines?

CHAPTER - 2

DC machines Generators



Learning objectives

Dynamically induced and statically induced e.m.f.
Working of single turn generator Fleming's right hand rule
Parts of d.c. generator
Lap and wave windings
E.m.f. equation of d.c. generator Separately excited generator Self excited

E.m.f. equation of d.c. generator Separately excited generator Self excited generator
Compound generator
Magnetization characteristic Internal and external characteristic

353

2.1. Introduction

D.C. machines are primarily d.c. generators and d.c. motors. We will study d.c. generators in this chapter and d.c. motors in the next chapter.
A generator is a machine which converts mechanical energy (or power) into electrical energy (power). Faraday's laws of electromagnetic induction state that
• When the flux linking with a conductor changes, e.m.f is induced in it.
• The e.m.f induced is proportional to the rate of change of flux.

$$e \propto \frac{Nd\varphi}{dt} \quad (2.1)$$

where e is e.m.f. induced
N is number of turns
φ is flux linking with conductor.

Lenz's law states that the e.m.f. is induced in a direction to oppose the cause producing it. The flux linking with a conductor can be changed by

• relative motion between conductor and magnetic field (producing the flux). This can be achieved by having a stationary field and rotating conductor or having a stationary conductor and rotating field.

• producing an alternating flux by an alternating current.

The machines where e.m.f. is induced using relative motion between conductor and field are called generators. The e.m.f. so induced is called "*motional e.m.f*"or"*dynamically induced e.m.f*". The e.m.f induced in a coil when an alternating flux links with it is called "*statically induced e.m.f*" and is used in transformers.

Thus in generators, the conductor cuts magnetic flux, so that dynamically induced e.m.f is produced in it. If the conductor circuit is closed a current flows in it. Hence, the essential parts of any generator are

• A magnetic field

- A magnetic field
• A conductor which moves relatively with respect to the field.

2.2. Principle of working

The principle of working can be best explained by considering a single-turn rectangular coil made of copper rotating about its own axis in a magnetic field provided by either permanent magnet or electro magnets. Consider a copper coil, ABCD, rotating in a magnetic field as shown in Fig. 2.1.

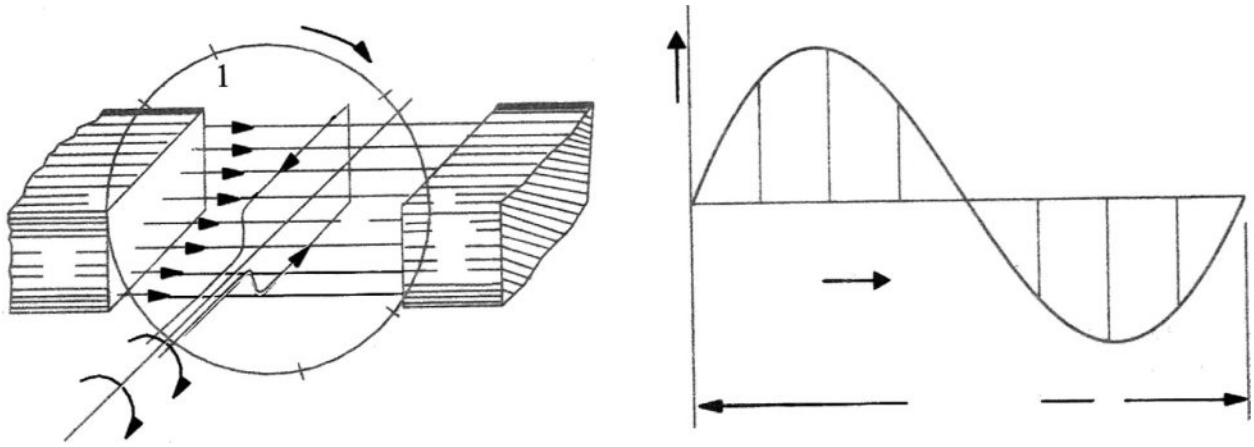
The two ends of the coil are joined to two copper rings called slip-rings, ‘a’ and ‘b’ which are insulated from each other and from the central shafts. Two brushes, of copper or carbon, press against the slip-rings. These brushes collect the current from the coil and take it to the external load resistance R_L.

Consider the position of the coil as shown in Fig. 2.1, as the reference. In this position, the plane of the coil is at right angles to the magnetic lines of flux. In this position, the rate of change of flux is minimum and no e.m.f is induced in the coil. This position is taken as the reference position. We measure the angle of rotation or time from this position. As the coil rotates, the rate of change of flux and hence the induced e.m.f increases, till at $\theta = 90^\circ$, when the coil plane is horizontal, the induced e.m.f is maximum.

In the next quarter, from$\theta = 90^\circ$ to 180°, the rate of change of flux linkages gradually decreases, till at $\theta = 180^\circ$ it is zero. Hence, in this position e.m.f is zero.

In the second half of the revolution, from $\theta = 180^\circ$ to$\theta = 360^\circ$, the variations of e.m.f are similar to those in first half of the revolution. It is minimum when plane of coil is perpendicular to lines of flux and maximum when the plane of coil is parallel to the lines of flux. If we consider the coil side AB we can see that in the first half of the revolution, it moves downward in the magnetic field, cutting the lines of flux. However, in the second

0°
8 A^2
270° B 3 12 3 4 **N**



D
56 7 8 9
S Time C_{46}
$5^{(180°)}$ Cycle
Figure 2.1 *Single turn coil.*
e.m.f.$^{\varphi m}$ 180° 270° 360°$_{0° 90° \theta}$
One cycle **Figure 2.2** *Induced e.m.f.*
half of the revolution, it moves upwards cutting the lines of flux. Therefore the e.m.f. induced in the coil is reversed. The induced e.m.f. is shown in Fig. 2.2.
The e.m.f induced in the conductor is given by
$e = B\vartheta \sin\theta$ (2.2) where e = induced e.m.f in volts

B = magnetic field intensity in T
= active length of conductor in m
ϑ = relative velocity between field and conductor in m/s
θ = angle between plane of rotation and plane of flux.

The direction of the induced e.m.f is found from Fleming's Right Hand rule. The application of the rule is shown in Fig. 2.3.

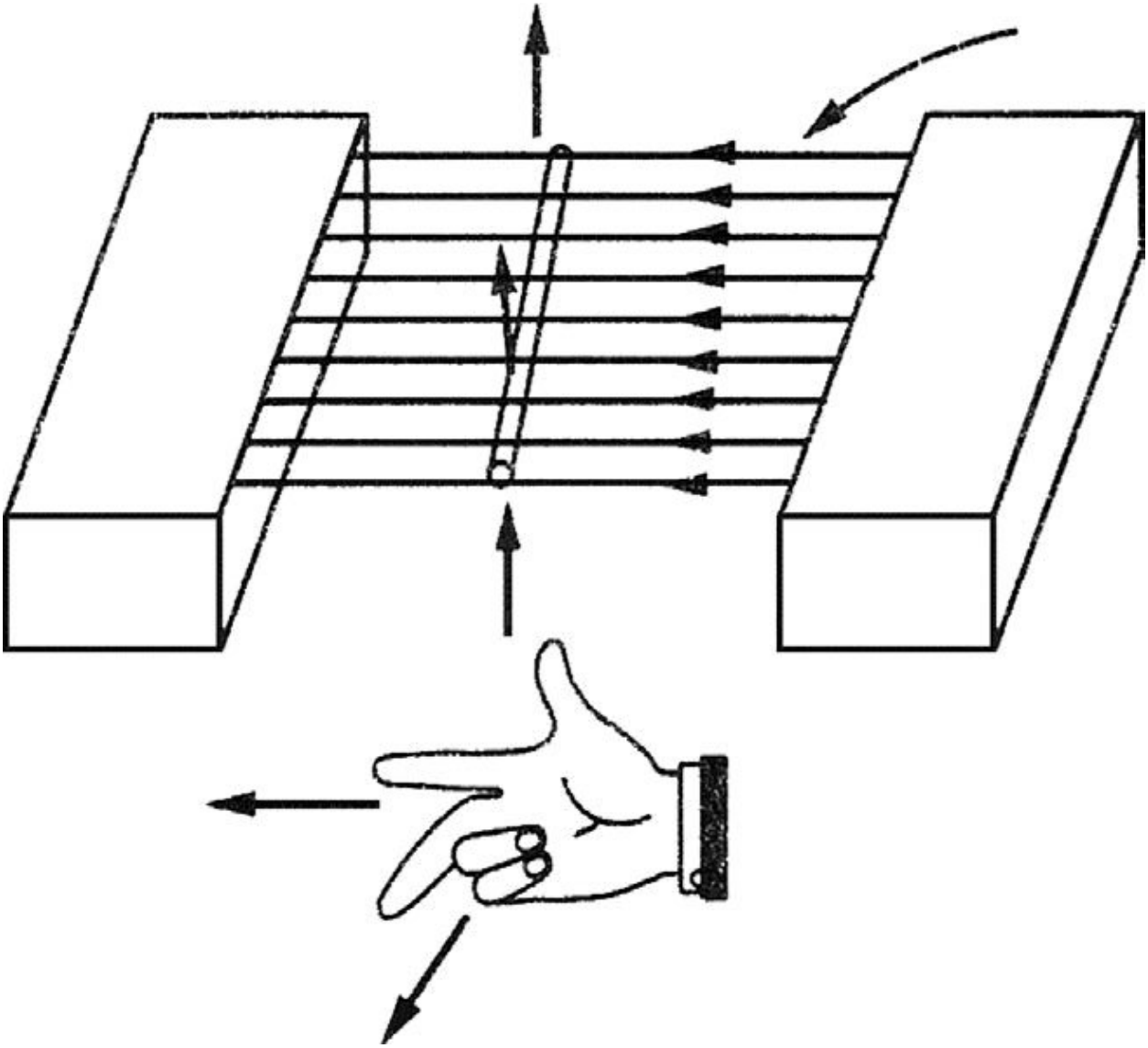
The thumb, fore finger and middle finger of the right hand are extended so that they are at right angles to each other. If the fore finger points in the direction of the lines of flux, the thumb in the direction of the relative motion of the conductor and the flux, the middle finger gives the direction of the induced e.m.f.

From Fig. 2.2, it is seen that the e.m.f induced in the coil is an alternating one. To convert it into a d.c. voltage we use a commutator. We next consider a practical d.c. generator and its construction.

2.3. Construction of a d.c. generator

While the single loop generator explained in the last section, illustrates the principle of the d.c. generator, the practical generator is shown in Fig. 2.4.

Motion of Lines conductor of flux



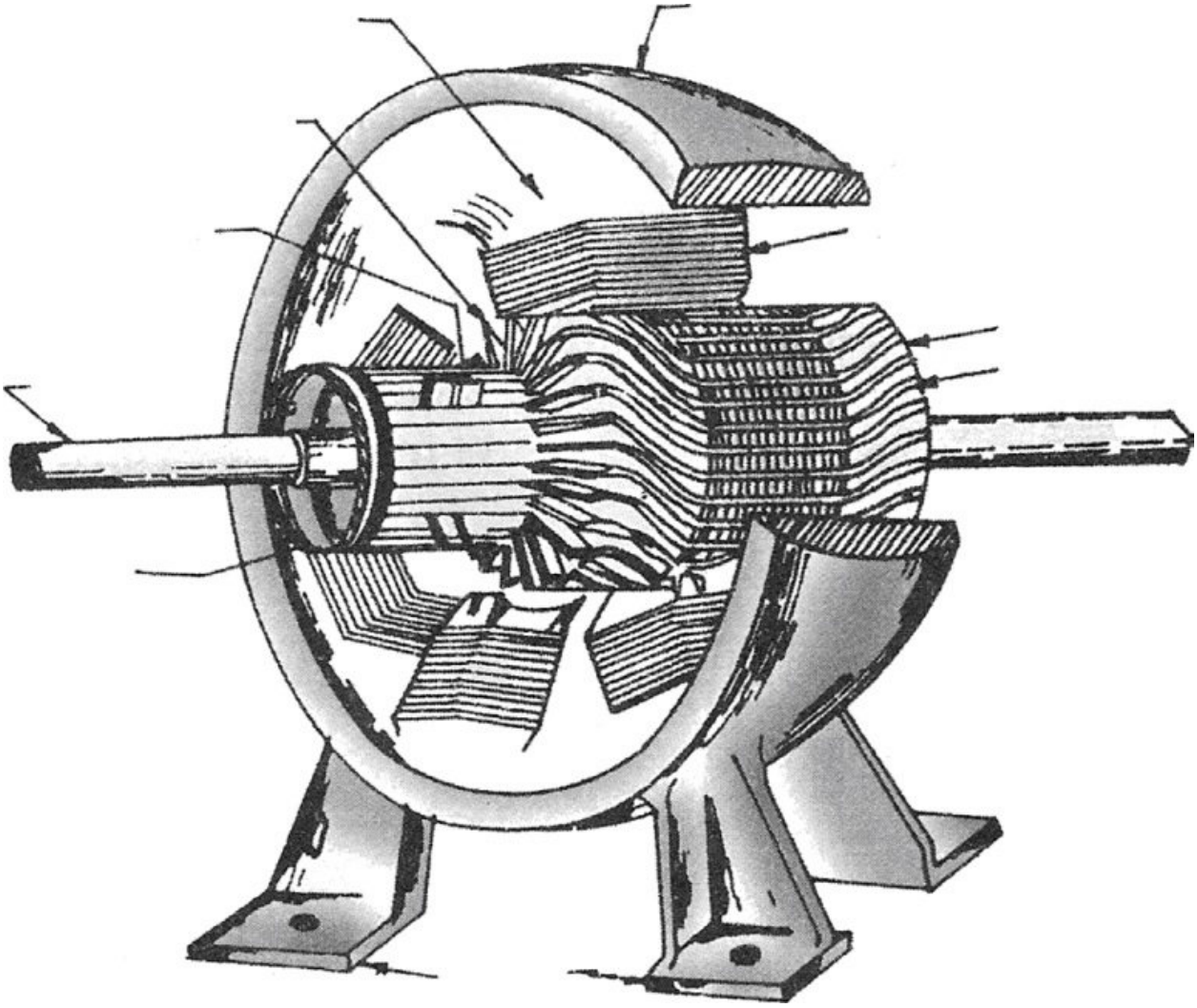
SNMotion
Direction
of flux
Direction of e.m.f. (current)

Figure 2.3 *Fleming's Right Hand rule.*

Field poles Yoke Lugs

Brushes
Field winding

Armature Shaft Conductors
Commutator



Feet
Figure 2.4 *D.C generator.*
The various parts of the d.c. generator are

1. Magnetic frame or Yoke
2. Armature core
3. Armature conductors or windings
4. Pole-core and pole shoe
5. Field coil (winding)
6. Commutator
7. Brushes and bearings.

The functions of the various parts are discussed below.

2.3.1. Yoke

The Yoke is the outer frame of the machine. The functions of the Yoke are

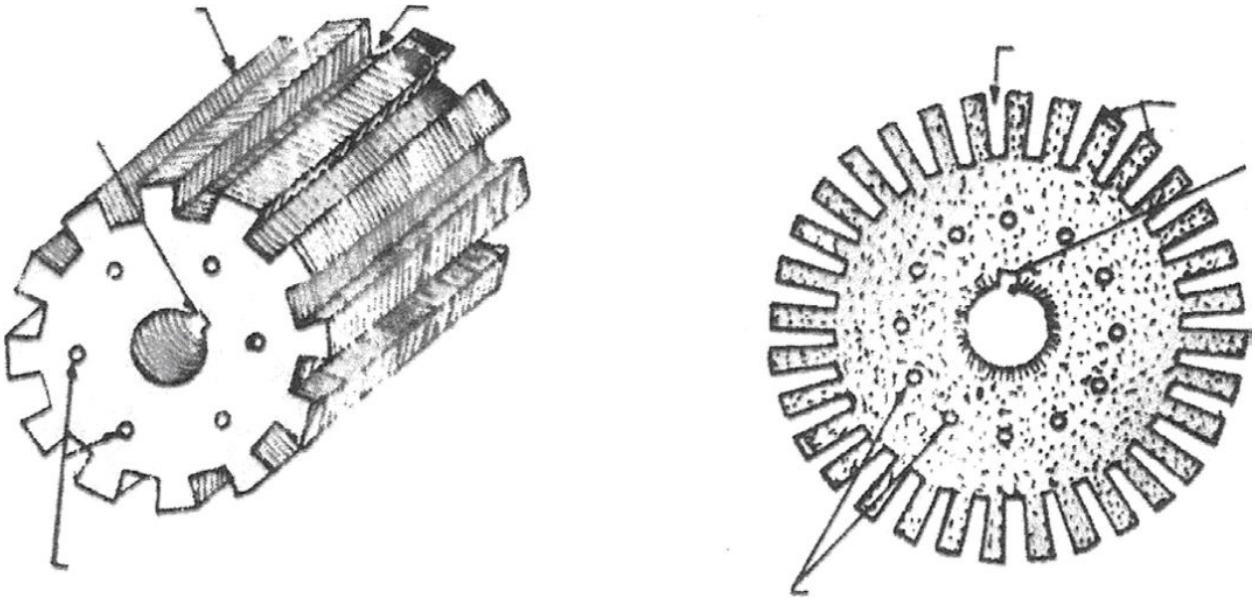
• To provide mechanical support to the poles.
• Act as protective cover for the machine.
• Carry the magnetic flux produced by the poles.

Yokes are made of cast iron because it is cheapest. For larger machines it is made of cast steel or rolled steel.

2.3.2. Armature core

The armature core is cylindrical in shape and is mounted on the shaft. It consists of slots on its periphery to house the conductors. It is built up of circular sheet steel discs or laminations around 0.5 mm thick. The core is keyed to the shaft. The armature core is shown in Fig. 2.5.

Laminators Slot$_{\text{Slot}}$



^{Teeth}Key way$_{\text{Key way}}$

Air holes$_{\text{Air holes}}$

Figure 2.5 *Armature core.*

The functions of the armature core are:

• To house the armature conductors
• To provide a path of low reluctance to the field flux from the N-pole to S-pole, through the armature.

It is made of cast iron or cast steel.

2.3.3. Armature windings

A number of conductors, housed on the armature core, are interconnected in a manner to add up the induced e.m.fs produced in the conductors. These interconnected conductors constitute the armature winding. The main functions of the armature winding are:

Generation of e.m.f•
• Carry the armature current supplied to the load in case of generators and carry the current supplied in case of d.c. motors.
The windings are basically of two types (i) Lap winding (ii) Wave winding. **Lap winding:** Here, if we start from the conductor in the first slot and number it as 1, the connections overlap as shown in Fig. 2.6.

The conductors are connected in series parallel combination to increase the voltage and current rating. Series connection increases the voltage rating and parallel connection increases the current rating.

In lap winding the total number of parallel paths is equal to the number of poles 'P'. Hence lap winding is preferred for machines with high current rating.
13 5
246
Figure 2.6 *Lap winding.*
13 5
24 6
Figure 2.7 *Wave winding.*
Conductor 1 Conductor 2
1 turn **Figure 2.8** *Turn in a winding.*
Wave winding: In this connection, the winding does not have overlap as shown in Fig. 2.7. Here the conductors get divided into two parallel paths irrespective of the number of poles. This type of winding is preferred for low current, high voltage generators. The following terms are commonly used with respect to windings.
• *Conductor:* It is the actual conductor placed in the armature slot. Each conductor is under the influence of either the north or south pole.

• *Turn:* Two conductors placed in different slots when connected together form a turn, as shown in Fig. 2.8.
The total number of conductors

Z = 2× number of turns (2.3) **Table 2.1** Lap and Wave winding.

Lap winding Wave winding

1. Number of Parallel paths, A = Number of poles, P.
2. Number of brushes is equal to P.
3. Preferred for high current-low voltage generators.
4. Used when current is greater than 500 A A = 2.

Number of brushes required is 2.

Preferred for low current-high voltage generators.
Used when current is less than 500 A

Yoke
Pole core
Pole shoe
Figure 2.9 *Poles.*

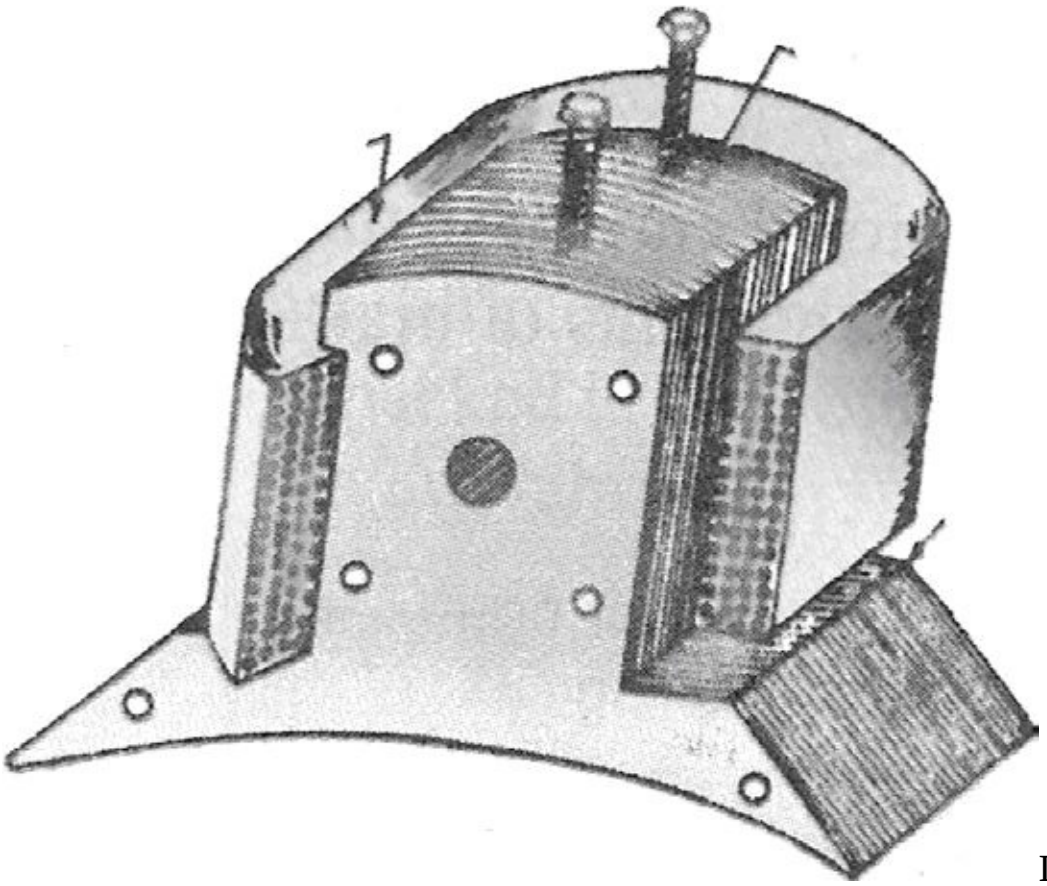
• *Coil:* The turns are grouped together to form a coil. If the coil contains only one turn its called a *single turn coil* and if it contains more than one turn its called a *multiturn coil*.

The comparison of lap and wave winding is given in Table 2.1.

2.3.4. Pole Core and Shoe
Each pole is divided into pole core and pole shoe as shown in Fig. 2.9.
The functions are as follows:

The pole core carries a field winding which is necessary to produce the magnetic flux. •
• It directs the flux through the armature core.

• The pole shoes spread out the flux in the air gap and being of larger cross-section, reduce the reluctance of the magnetic path.
Laminated $_{\text{Exciting}}$ pole core coils



Laminated pole shoe

Figure 2.10 *Field coil.*

In modern designs, the pole core and shoes are both built of thin laminations of annealed steel, of thickness between 0.25 mm to 1 mm. The pole is then securely bolted to the yoke.

2.3.5. Field coil (winding)

The field coils (or pole coils) are made of copper wire or strip and are former wound. The wound coil is put into place over the pole core as shown in Fig. 2.10.

The function of this coil is to produce the necessary flux, when current is passed through it. Since it produces the flux its also called the *exciting winding*. The field coils are so connected that they form alternate N and S poles.

2.3.6. Commutator

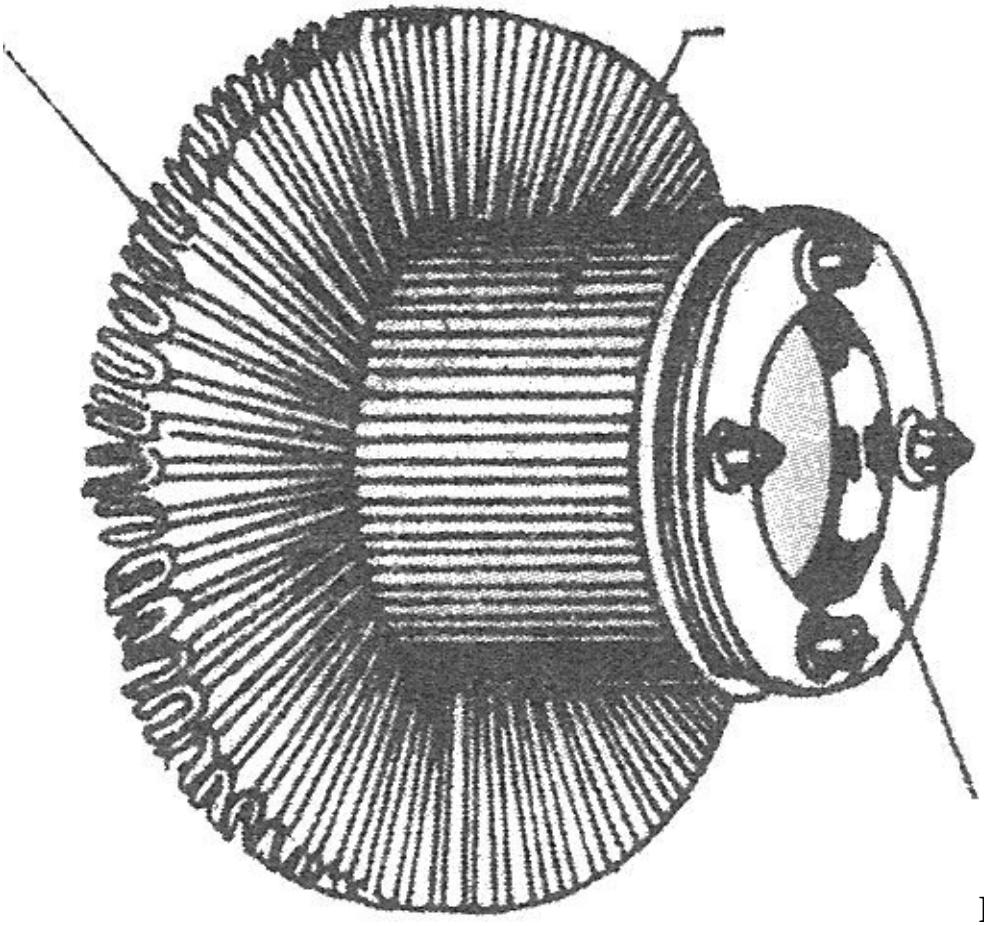
The basic nature of e.m.f. induced in the armature conductors is a.c. as discussed in section 2.2. The commutator is the device which rectifies it to d.c. It facilitates the collection of current from the armature conductors. It is of cylindrical shape and built up of wedge shaped segments of high conductivity of hard drawn

and built up of wedge-shaped segments of high-conductivity of hard drawn copper. The segments are insulated from each other by thin layers of mica. Each segment is connected to the armature conductor by means of a copper strip or plug. The commutator is shown in Fig. 2.11.

2.3.7. Brushes and bearings

Brushes are stationary and rest on the surface of the commutator. Their function is to collect current from commutator and make it available to the external circuit. They are

Commutator Insulted
lugs
copper
segments



End
clamp

Figure 2.11 *Commutator.*

made of a soft material like carbon. They are housed in box-type brush holders. The brushes are made to press on the commutator by a spring whose tension can be adjusted. A flexible copper pigtail mounted at the top of the brush conveys

be adjusted. A flexible copper pigtail mounted at the top of the brush conveys current from the brushes to the holder.

Ball bearings are used commonly because of their reliability. Roller bearings are used for heavier duties.

2.4. E.m.f. equation of d.c. generator

Let φ = flux/pole in Wb
Z = total number of armature conductors
P = number of poles
A = number of parallel paths
N = speed in revolutions per minute (rpm)

E_g = induced e.m.f.

Average e.m.f 'e' generated per conductor = $\frac{d\varphi}{dt}$. The flux cut by the conductor in one revolution

dφ = φP W b (Since φ is the flux/pole,φP gives total flux)

Number of revolutions per second = $\frac{N}{60}$

∴. Time taken for one revolution = $\frac{60}{N}$

Hence,

e

=

$\frac{d\varphi}{dt} = \frac{\varphi P}{60} = \frac{N\varphi P}{60}$

Number of parallel paths = A
Number of conductors/path
=
Z
A
∴
E.m.f generated/path

=

$Z \frac{N\varphi P}{60} = \frac{ZN\varphi P}{60A}$

- A = P, for lap winding
- A = 2, for wave winding

• Z = No. of armature slots× No. of conductors/slot

$E_g = \frac{ZN\varphi P}{60A}$ Volts (2.4)
Since Z, P, and A are constants,
$E_g = K_a \varphi N$ Volts (2.5)

Example 2.1 A 4 pole, lap wound d.c. generator has flux of 0.075 Wb/pole. What is the e.m.f. generated when it is rotated at a speed of 1000 rpm, if the number of armature conductors is 440? What would be the e.m.f. if the machine is wave wound?

Solution: P = 4; Z = 440; φ = 0.075; N = 1000 rpm. For lap winding A = P = 4
$E = \frac{440\times 1000\times 0.075\times 4}{60\times 4} = 550$ V.
For Wave winding A = 2

$E = \frac{440\times 1000\times 0.075\times 4}{60\times 2} = 1100$ V.
Example 2.2 An 8 pole d.c. generator has 500 armature conductors and a useful flux of 0.06 Wb/pole. What is e.m.f. generated if it runs at 1200 rpm and conductors are lap wound? At what speed must it be driven to produce the same e.m.f. if it is wave wound?

Solution: If it is lap wound A = 8
$\frac{500\times 1200\times 0.06\times 8}{60\times 8} = 600$ V. Eg = 60× 8
If it is wave wound, then A = 2
$\frac{500\times N\times 0.06\times 8}{60\times 2} = 600$ N = 300 rpm.

Example 2.3 A four pole generator having wave wound armature winding has 50 slots, each containing 20 conductors. If the flux per pole is 6.8 mWb and speed is
1500 rpm, find the voltage generated?

Solution: Z = No. of slots × conductors/slot = 50× 20 = 1000 $1000\times 1500\times 6.8\times 10^{-3}\times \frac{4}{60\times 2} = 340$ V. Eg = 60× 2

Example 2.4 A 4 pole lap-wound d.c. generator having 80 slots with 10 conductors/ slot generates 400 V when running at 1200 rpm. How can a voltage of 360 V be obtained from the same machine?

Solution:

$E_g = \frac{ZP}{} \varphi N = Ka\varphi N_{60A}$

$ZP = Ka = \frac{80\times 10\times 4}{60A} = 13.33.\frac{}{60\times 4}$

$E_{g1} = 400 = 13.33\times 1200\times \varphi_1$ $\varphi 1 = 0.025$Wb.

We can get 360 V by either keeping same flux and changing speed or by keeping the speed constant and changing flux.

Keep flux constant Then $E_{g2} = 360 = 13.33\times 0.025\times N$

N = 1080 rpm.

Keep speed constant Then $360 = 13.33\times \varphi \times 1200$

$\varphi = 0.0225$ Wb.

Hence, 360 V can be generated by changing the speed from 1200 rpm to 1080 rpm keeping the flux at 0.025 Wb or by changing the flux from 0.025 Wb to 0.0225 Wb keeping the speed at 1200 rpm.

2.5. Types of d.c. generator

All the generators need to have an excitation source, which is responsible for producing the magnetic field. Based on the way in which the field is classified, generators are classified into

- Separately excited
- Self excited

2.5.1. Separately excited generator

In these generators the field magnets are energized from a separate independent external source of d.c. current. This is shown in Fig. 2.12.

A separate d.c. source is used to drive the field current I_f. The resistanceR_f is connected to adjust the field current. The e.m.f. generated in the armature drives the load. It can be seen that the armature current is same as the load current.

2.5.2. Self-excited generator

When the electromagnets are deenergized, there is always some retention of magnetism in the poles. This flux is called *residual flux*. When the armature is now rotated, an e.m.f. is generated. This is used to drive the field current and strengthen the residual pole flux. There are three types of self excited generators.

I_f I_a
R_f +
D.C. Armature Load source$_{Fld\ -wdg}$
Figure 2.12 *Separately excited generator.*
R
$^{f\ If}$ I_a I_L +
$_{F1}$ A_1 + F_1−F_2: Field winding A_1−A_2: Armature$_{V\ Load}$
F
2
V: Terminal voltage $A_{2\ -}$
− **Figure 2.13** *Shunt generator.*

2.5.3. Shunt generator

In these generators the field windings are connected across the armature and the full voltage is applied to the field circuit as shown in Fig. 2.13.
The field winding has a large number of turns of thin wire. It has a high resistance,R_{sh}. The equations of the shunt generator are as follows:

Ia = If + IL (2.6)

$^{V\ (2.7)If\ =}$ Rsh + Rf

S_1 $S_{2\ If\ R}$ +
$f^{I}LI_a$
A
1
$_{+}$ S_1−S_2: Series field A_1−A_2: Armature$_{V\ Load}$
$A_{2\ -}$
− **Figure 2.14** *Series generator.*

(Note that R_{sh} is the resistance of the field winding and R_f is the external resistance connected)

Eg = V + IaRa + Vbrush (2.8)

In (2.8), E_g is the induced e.m.f, R_a is the resistance of the armature winding and V_{brush} is the voltage drop due to brushes, called the brush contract drop.V_{brush} is generally small and can be neglected. It is around 0.5 V for metallic brushes and 2.0 V for carbon brushes. The generated e.m.f E_g is also given by

$_{Eg\ =}$ ZNφP$_{.60\ A}$

2.5.4. Series generator